Long-term 2029 Feasibility Study

nationalgrid

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Contents

Authorisations

Name	Role Title	Department	Signature	Date Signed
Christopher Hartley	Regional Connections Manager	Customer and Network Development		21/10/2025

Version Control

Issue	Date	Summary of Changes
2.1	21/10/2025	Final
1.0	08/05/2025	Draft v 1.0

Executive Summary

National Energy System Operator (NESO) have commenced a procurement event to secure various network services named Long-term 2029.

To support the delivery of Long-term 2029, several connection points (bays) on the network have been pre-emptively reserved for the successful tender participant(s).

This exercise was undertaken following a high-level substation assessment exercise. This avoids the want / need for individual participants to submit their own connection application / modifications until the outcome of the tender is known.

NESO Note: *Please note in line with the connection requirements, bidders do not have to rely on any reserved bay and can choose to bid on the basis of their own connection agreement / offer should they hold one.*

National Grid Electricity Transmission Owner ("NGET") have undertaken this desktop feasibility analysis to provide a clear pack of information on the scope of works required to facilitate a connection onto the National Electricity Transmission System for each of the reserved bay sites to form part of the information pack that will be issued as part of the Long-term 2029 Invitation to Tender.

Limitation of Study

This technical assessment has been undertaken using a mixture of desktop technical tools and databases to provide a technical indication of the scope of works, costs and lead time required for new connections on NGET's transmission network. For the avoidance of doubt the study has not included any site visits, surveys, or similar.

The assessment and resulting final report are a snapshot in time and does not replace the formal CUSC connection process. It should be noted that final confirmation of any connection agreement is subject to the formal CUSC connection process. NGET cannot guarantee that a chosen site may be suitable or available at the time of the CUSC connection application.

Confidentiality

This document is being provided to NESO and the defined participants of the tender event for the Long-term 2029 tender on a strictly confidential basis and each recipient shall not communicate, reproduce or disclose any part of this document, without the express written consent of National Grid Electricity Transmission plc to do so and any unconsented act may constitute a breach of confidentiality.

Aims and Assumptions

NGET will undertake a desktop only assessment for NESO's Long-term 2029 tender. That is, no site visits or site based engineering or geotechnical surveys, will be conducted as part of the assessment. The study will cover two aspects: a substation site study and a power systems feasibility assessment.

The assessment and final report, including NGET's indicative programme and cost estimates for any triggered infrastructure / one-off works, are a snapshot in time and does not replace the formal CUSC connection process.

The aim of the technical assessment is to provide:

- An overview of the proposed connection site, information on the connection option available and high-level site knowledge, including indicative cost information and connection dates.
- System studies will look at voltage step compliance, fault-level analysis and provide information on the availability of any thermal MW headroom.

Assumptions

- Standard CUSC connection application process applies when the bidder applies for any reserved bay following success in the tender (both NESO and NGET acknowledge this will likely be the new reformed process following Connections Reform)
- Any connection applications or offers already in flight prior to NESO reserving any bays will be included in the assessment as ahead in the queue (though noting that the G2tWQ process is upcoming based on Connections Reform)
- All requirements and obligations from Grid Code, CUSC, NETS SQSS will apply unless otherwise specified
- NGET will assume only one tenderer will proceed per reserved bay.
- NGET will assume this will likely be a 0MW synchronous compensator, a 0MW reactor or a BESS asset. NGET will not, at this stage, consider a wider range of technology options at each site.
- NGET assessment will be based on infrastructure assets only and the standard CUSC ownership boundaries will apply. It is assumed that the User will own and operate all their assets. However, when going through the connections process a User may request NGET deliver connection asset works at their own discretion. In the process of the study work should NGET identify any work that could become connection asset works, an indicative cost estimate will be provided by NGET for this work in the report. This should be clearly communicated by NGET in the report with the rationale followed to calculate indicative costs.
- Any User asset costs will need to be considered and costed by the Tenderer.
- NGET assumes all User works will be delivered by the Tenderer who will seek and ensure they have all necessary consenting rights, permits, land and access rights, subject to the general information regarding non-operational land provided by NGET.
- **All costs estimated during the studies are indicative and are subject to change**.
- All connections are subject to availability of outages and resources, NGET will check outage availability as part of the assessment, but this cannot be reserved. Outages will be finalised upon the receipt of any connection application as part of the usual connections process, following the Long-term 2029 tender.
- **Non-operational land is not reserved as part of any bay reservation or as result of this report.**
- ‘Earliest In-Service Date’ (EISD) is to be assessed against ‘NGET TP500 Runways v1.5’ standardised investment programs, with reasonable float included.
- NGET Investment/Construction program based on initial application for connection in circa ~ June 2026 (though note this is subject to the next available connections application window following the completion of the Long-term 2029 tender).
    - Note: any delay will result in a slippage of the forecasted EISD by at least same duration if not more due to network access period during any Winter months. There may be opportunities to address any delay or even accelerate the NGET works, but this can only be understood and quantified once a competent connection application is made, and the necessary design development and site investigation work have been undertaken.
- NGET should assume that the connection required is a Firm connection as per the SQSS compliant connection design. In areas where there are MW restrictions, non-firm connections should be facilitated, enabling providers to connect their solutions early and provide MVAr/fault-level support whilst waiting for any reinforcements to be delivered to export MW.

Substation Site Review

Ref	Location of Bay Reservation (Substation)	No. of Bays	Initial Bay Reservation – Prior to detailed feasibility studies (Bay Identifier)	Final Bay Reservation Confirmation – Following completion of feasibility studies (Bay Identifier)
1	Greystones A/B 275kV	1	1 x bay transfer. Existing bays are installed at these sites, and the view is these will be transferred to the 'USER' to refurbish.	It is recommended that the service provider user is allocated an old power station generator bay as the Point of connection. This bay will need to be refurbished by the service provider and made suitable for their purposes and compliant with all NGET policies.
2	Penwortham 400kV	1	New substation, as such a new bay.	The new 400kV GIS substation will have space for additional bays on either side of the bus coupler. A reserved bay connection will be available circa 2034, but this is dependent on the new GIS substation build progressing. Otherwise, Penwortham has no place for any additional connection.
3	Templeborough 275kV	1	1 x reserved bay will be based on the reconfiguration of Mesh Corner 4.	Reserved bay will be based on the reconfiguration of Mesh Corner 4 and removal of the old transformer plinth, bund and associated civil structures to create space for a NGET owned connection bay for the service provider.
4	New High Marnham 400kV	1	New substation, as such a new bay.	No exact bay identified as new sub is being delivered by SI, however, they have made a provision for a future bay. Substation is due to be completed circa 2029 with circuit transfer by circa 2031.
5	Ocker Hill 275kV	1	1 x new bay based on the reconfiguration of the substation. However, this is not a recommended location by NGET for LT29 schemes.	The conclusion on Ocker Hill is that to support 1 x extra bay at the site NGET would need a decision as to if LT29 solution is deemed feasible for the site and support will be needed to manage the affected BESS customer. The work involved would firstly be expensive and extensive to support the two connections, therefore this site is not a recommended location by NGET for LT29 schemes.
6	Drakelow 400kV	1	1 x new bay as part of the site rationalization.	This site is a possible location for an LT29 scheme, it is worth noting that access to the site is limited and constrained. A LT29 scheme can be facilitated at this site around 2030-2031.

7	New Kemsley 400kV	1	New substation, as such a new bay.	North Kent Connection Node A is a new substation that will be built close to the existing Kemsley substation. Final location TBC pending siting study with an indicative preferred location to be identified in spring 2026. The new substation will reserve a spare bay for the LT29 scheme.
8	Richborough 400kV	2	2 x new bay at Richborough 400kV	There is space available within Richborough 400kV GIS substation for two bays (2 x bays) to extend the GIS hall to the west for the pathfinder bay. NGET will undertake the GIS hall extension work with the customer responsible for procuring and installing the GIS bay.
9	Landulph 400kV	2	2 x new bay at Landulph 400kV	The LT29 scheme to facilitate two bays (2 x bays) will consist of a single busbar and separate single-switch mesh arrangement made at Busbar 3 adjacent to Langage circuit 2. Adjacent external land acquisitions, land works and cable easements would need to be determined by the customer.
10	New Woolavington 400kV	1	New substation, as such a new bay.	The new 400kV GIS substation will have space for additional bays. A reserved bay connection will be available circa 2030, but this is dependent on the new GIS substation build progressing. Otherwise, Woolavington 400kV has no place for any connection.

Greystones A & B 275kV

Reserved Bay Request

The original request was made for Lackenby 400kV substation, however Lackenby is full and cannot be extended, Greystones 275kV A & B substations (both indoor GIS technology) are nearby adjacent alternative sites suggested by NGET as an alternative to Lackenby.

There are around 8-10 old ‘spare’ power station bays at this site, but some will be needed for contracted customers. As such, exactly which of these bays will be allocated needs to be finalised.

Note: Greystones A&B are old ex power station substation sites, the spare CB’s and bays have not been maintained since the old power station was decommissioned, the CB and bays will need to be refurbed as part of the USER connection works.

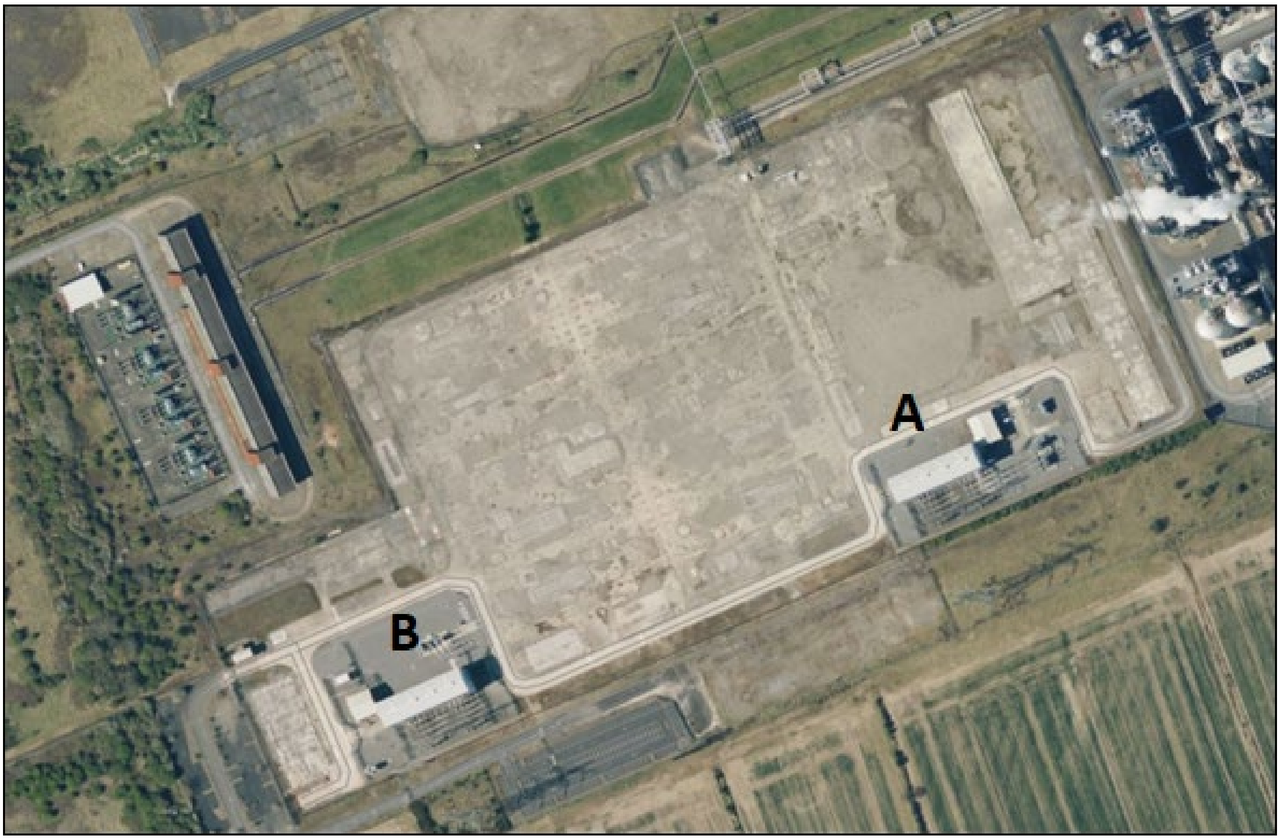
The remaining report provides more details about the reserved bay following the original bay reservation request.

Review of HV Bay Connections

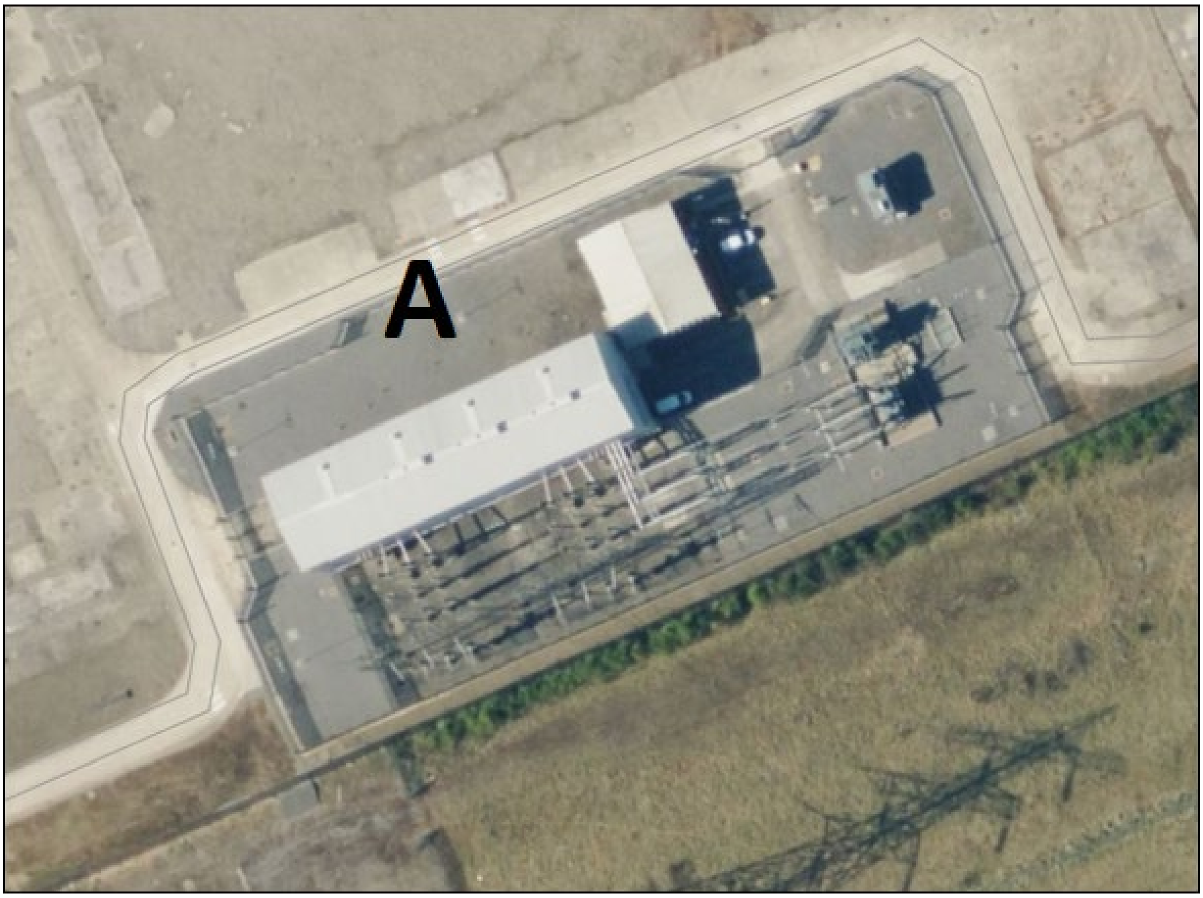
Greystones A & B are both indoor GIS substations.

The substation(s) were constructed and commissioned circa 1992.

Overview:



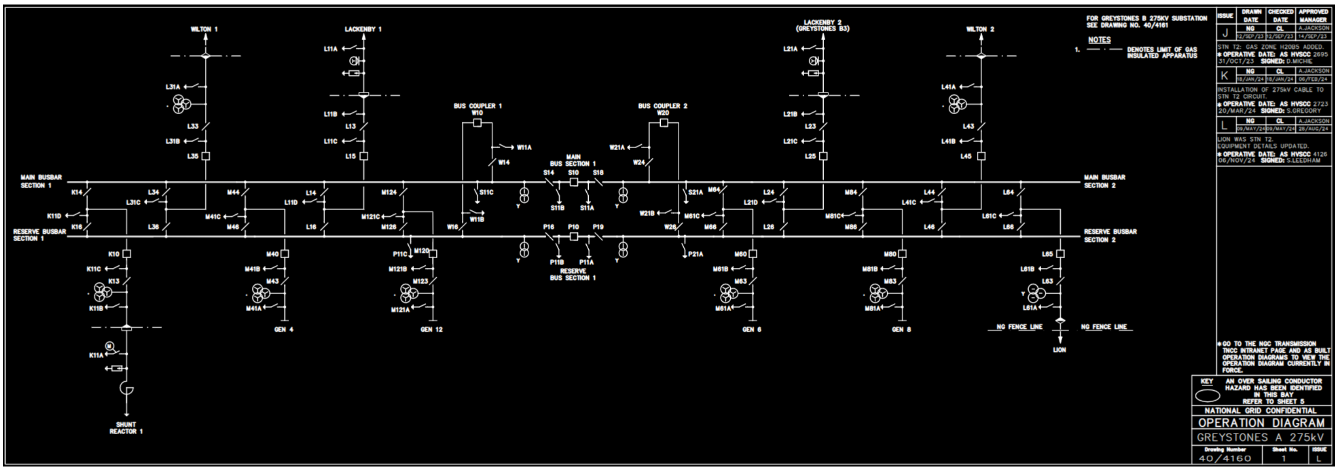
Greystones A:



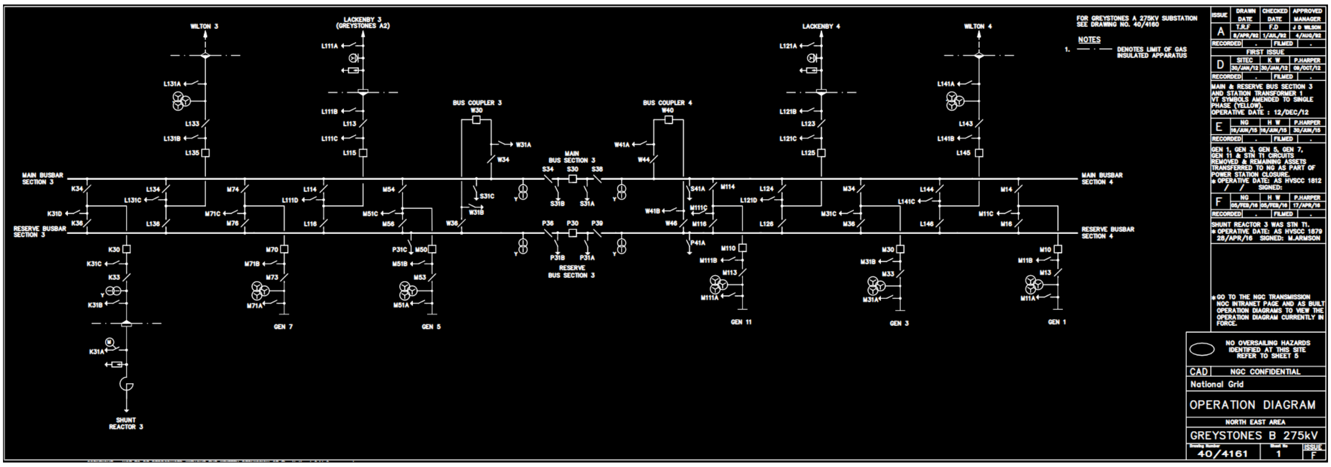
Greystones B:

Running Arrangement diagram

Greystones A:



Greystones B:



Indicative Completion Date for Connection

Scope of Work	1. GIS Bay transfer (1992 existing generator bay to be refurbished by the USER, if this is not suitable or acceptable the USER will need to provide a 'new' USER connection bay). 2. Commissiong and database changes as required. 3. Any mods to the building, civil structures and P&C requirements. Note: The condition of the redundant bays will need to be evaluated by the service provider for their use.
Completion of NGET works (Indicative)	31st October 2030 but will need to align with the USER refurbishment works and commissioning programme.
Indicative Connection / EISD (Indicative)	31st October 2030 but will be determined by the appropriate delivery team during scheme development.

Note: The working assumption is the application for the connection would be made in / around ~ June 2026 or the next available application window to achieve the EISD.

Review on Outage Availability and Resource

No outages or resources have been secured at this time due to the uncertainty of the connection program.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	£250k
Connection costs (if applicable) *	None
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£300k**
Total Cost (With optional Connection Costs) *	**£300k**

Note 1: This is based on the re-use of the existing bay equipment.

Note 2: If a new bay is required this scope will need to be revisited and new equipment installed. The costs for new equipment will be significantly higher.

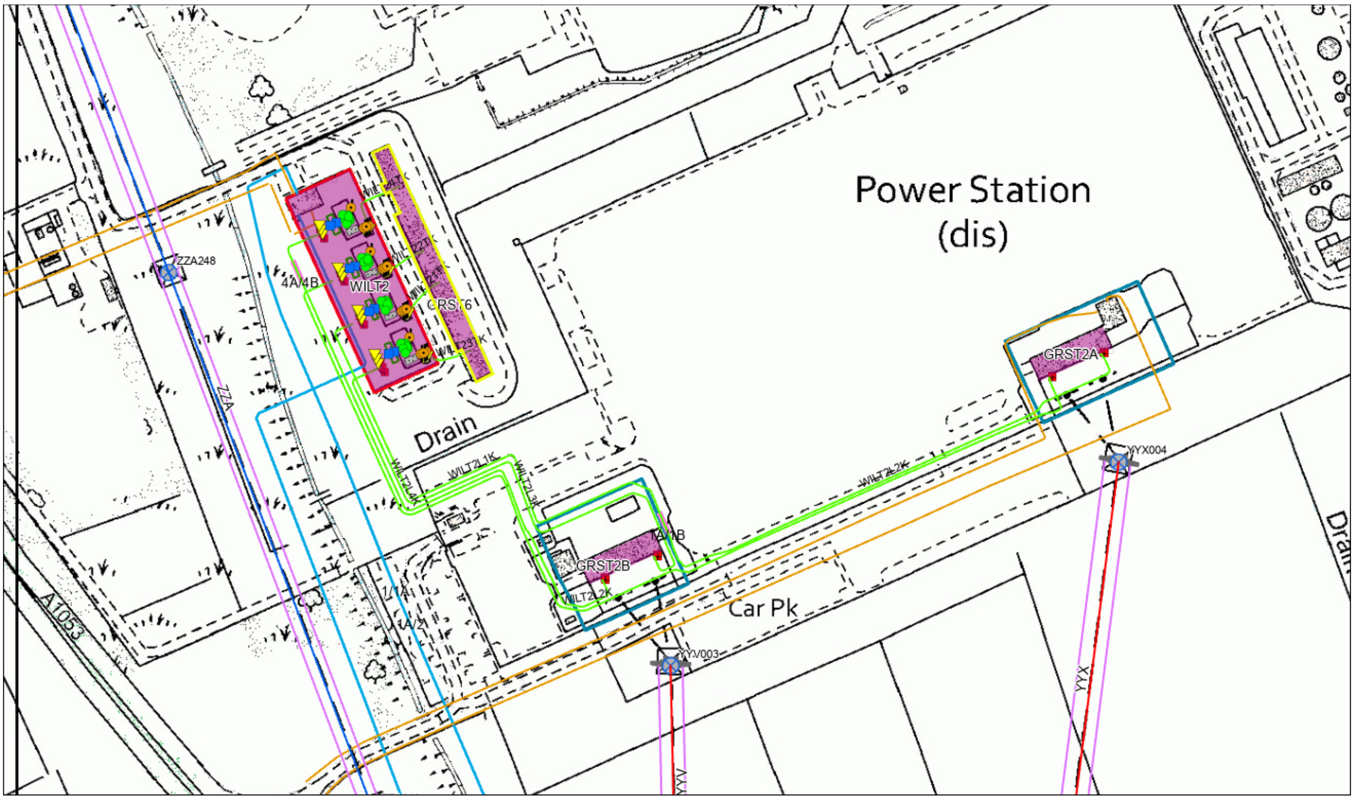
Note 3: Any impact on costs requiring either ARS or PoW are as follows:

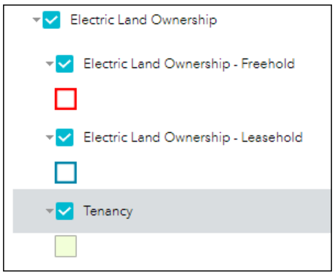
Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

Land Ownership:

Greystones A & B land ownership (it is believed that NGET are tenants at Greystones, we don't own any land):





Site Info	Response
Technical Limitation (TL's)	No
Risk Management Hazard Zones (RMHZ)	No
Operational Engineering Safety Bulletins (OESB)	No
ISS classification / security	Standard Security measures at present but may change to ISS.
Flood Risk	No
Additional Site Information	Shared access road with 3rd party landowner.

Load/Connection Offer Interactions

100158 - GRST2 Gen Conn of 340MW BESS Sembcorp, ACL 30/11/2025
100673 - GRST2 TEES CCPP 850MW BESS, ACL 30/09/2026
102527 - GRST2 Generation Connection of 299MW OCGT Sembcorp Utilities Limited, ACL TBC

Non-Load Interactions

102828 - SF6 Emission Abatement -Greystones 275kV, ACL 31/03/2031
104344 - YYV GREYSTONES – LACKENBY, ACL TBC

High Level Delivery Programme for Greystones A & B 275kV

	2026				2027				2028				2029				2030			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																				
ESO Tender																				
Optioneering and Developemnt																				
Sanction / Tender and Construction																				
Commissiong																				
Closure																				
BP 500Milestones (Typical program)	A0		A1		Gate B			Gate C			FSA				ACL		Closure			

Summary & recommendation

It is recommended that the service provider user is allocated an old power station generator bay as the Point of connection. This bay will need to be refurbished by the service provider and made suitable for their purposes and compliant with all NGET policies.

At this time the exact bay the service provider is allocated will need to be determined as part of any connection application (it could be at this time in Greystones A or B 275kV indoor GIS substation). There are several other connections at these sites and the exact bay allocation for multiple USERs will be coordinated as part of the scheme progression.

However, one of these bays will be kept reserved and available for any service provider.

NGET scope is to provide Commissiong and database changes as required and any modifications to the building, civil structures and P&C requirements.

Penwortham 400kV

Reserved Bay Request

The original bay reservation request was made by NESO based on the following information:

NGET are building a new GIS 400kV substation in the footprint of the existing 275kV operation area, due to be completed early to mid-2030's. It was also advised that NGET need to build interconnection between existing AIS 400kV and new GIS 400kV compounds. The new GIS substation will have space for the additional bays on either side of the bus coupler.

There is an opportunity for a bay within this, but the design will need to be confirmed, and the bay allocated as part of the new substation optioneering and design phases.

The remaining report provides more details about the reserved bay following the original bay reservation request.

Review of HV Bay Connections

Location of the new 400kV GIS substation and bays TBC.

Running Arrangement diagram

Not available as the new substation layout and design is not completed at this time.

Indicative Completion Date for Connection

Scope of Work	The new GIS substation we will have space for the additional bays on either side of the bus coupler. Dependency: only available if the new GIS substation is built, otherwise Penwortham has no place for any additional connection.
Completion of NGET works (Indicative)	31st Oct 2034.
Indicative Connection / EISD (Indicative)	31st Oct 2034 (However, if the new GIS substation build can be accelerated there is a possibility the EISD date could be earlier. This is not confirmed at this time but will be discussed during scheme development).

Note: The working assumption is the application for the connection would be made in / around ~ June 2026 or the next available application window to achieve the EISD.

Review on Outage Availability and Resource

None available currently.

The new substation will be an offline build, as such outages should be available, resources will need to be confirmed as part of any scheme progression.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	£5,500k
Connection costs (if applicable) *	£5,500k
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£5,550k**
Total Cost (With optional Connection Costs) *	**£11,050k**

Note 1: As this is a GIS new build, it may be pragmatic to include the option of a Connection Asset to avoid Tiger bars and multiple contractors working concurrently to gain build efficiencies. The offer of connection assets will be determined at customer application on a case by base review.

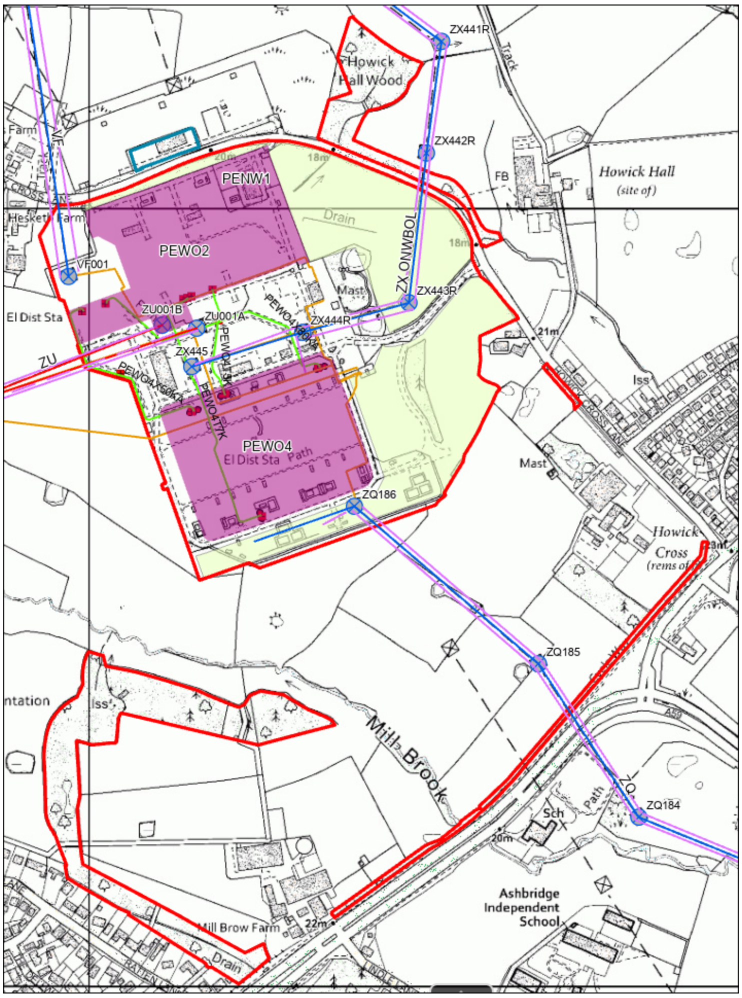
For clarity, the minimum costs for this connection, without the optional connection costs, would be £5.55M. If the optional connection costs are elected, this £5.55M will increase by £5.5M resulting in £11.05M.

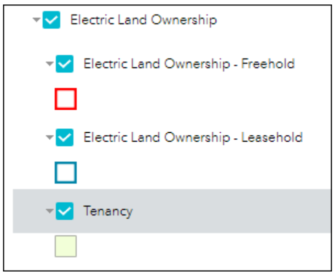
Note 2: Any impact on costs requiring either ARS or PoW are as follows:

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

Land Ownership:





Site Info	Response
Technical Limitation (TL's)	Yes
Risk Management Hazard Zones (RMHZ)	Yes
Operational Engineering Safety Bulletins (OESB)	Yes
ISS classification / security	Penwortham 400kV substation has ISS status and Cyber secure access installed
Flood Risk	Yes
Additional Site Information	Penwortham 275kV has Smart wires installed

Load/Connection Offer Interactions

101569 - PEWO4 Innova 600MW BESS, ACL 30/10/2030

101607 - PEWO2 Shaw Energi 57MW SGT1B BESS - CON, ACL 31/08/2026

101946 - PEWO4 BESS CSE28L 57MW SGT12 Tertiary, ACL 01/06/2027

102352 - PEWO4 ENWL New SGT in eastern extension, ACL 30/10/2036

102711 - PEWO4 Morecambe INFRA HND, ACL 27/11/2028

102712 - PEWO4 Morgan 1500MW Wind, ACL 30/11/2029

Non-Load Interactions

100476 - VF 058-071 HEYS-PEWO-STAH OHL Full Refurb, ACL 31/10/2025

103964 - PEWO Holistic Site Strategy, ACL 30/10/2033

104167 - PEWO Holistic Site Strategy - MSC 1, ACL 31/10/2028

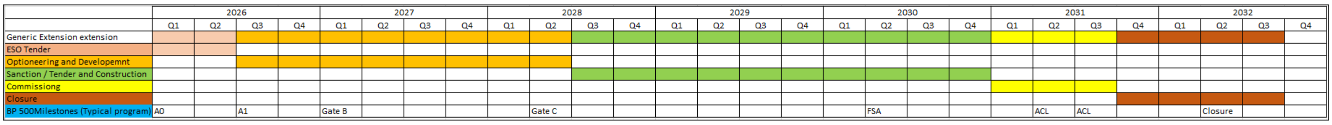
104447 - PCR1 - reconductor CARR-PEWO, PADI-PEWO, ACL TBC

104477 - PEWO Holistic Site Strategy - MSC 2, ACL 31/10/2029

101915 - Carrington - Penwortham ZQ OHL Uprating, ACL 25/08/2028

102368 - Penwortham extensions, ACL 30/10/2026

High Level Delivery Programme for Penwortham 400kV



Note, this is a generic delivery program.

The new Penwortham substation is due to be completed circa 2033 with an EISD of 2034, as such the years on this program need to be adjusted accordingly pro-rata to the new substation build.

Summary & recommendation

The new 400kV GIS substation will have space for additional bays on either side of the bus coupler to facilitate a reserved bay.

A reserved bay connection will be available circa 2034, but this is dependent on the new GIS substation build progressing. Otherwise, Penwortham has no place for any additional connection.

The exact bay allocated for the service provider will be determined as part of the new substation optioneering and design process.

Additional discussions will take place with the service provider as part of the substation design process to factor in the service providers requirements and an agreement will be reach between all parties to confirm the bay allocation for the service providers Point of Connection.

Templeborough 275kV

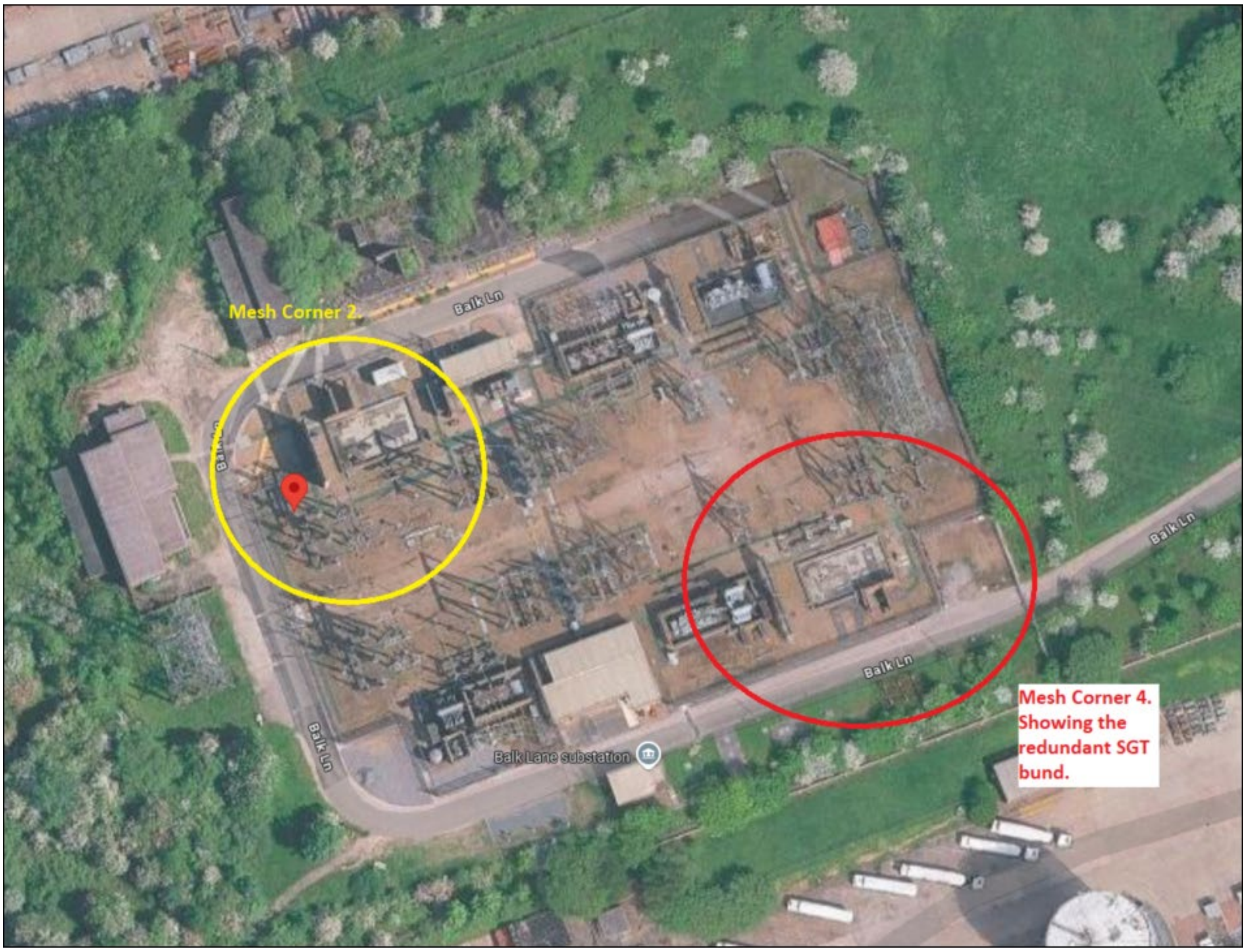
Reserved Bay Request

There is the possibility of reconfiguring MC4 and removing the old SGT4 infrastructure to create a Point of Connection. This will rely on a busbar extension and work to repurpose the old transformer plinth and bund to create the Point of Connection.

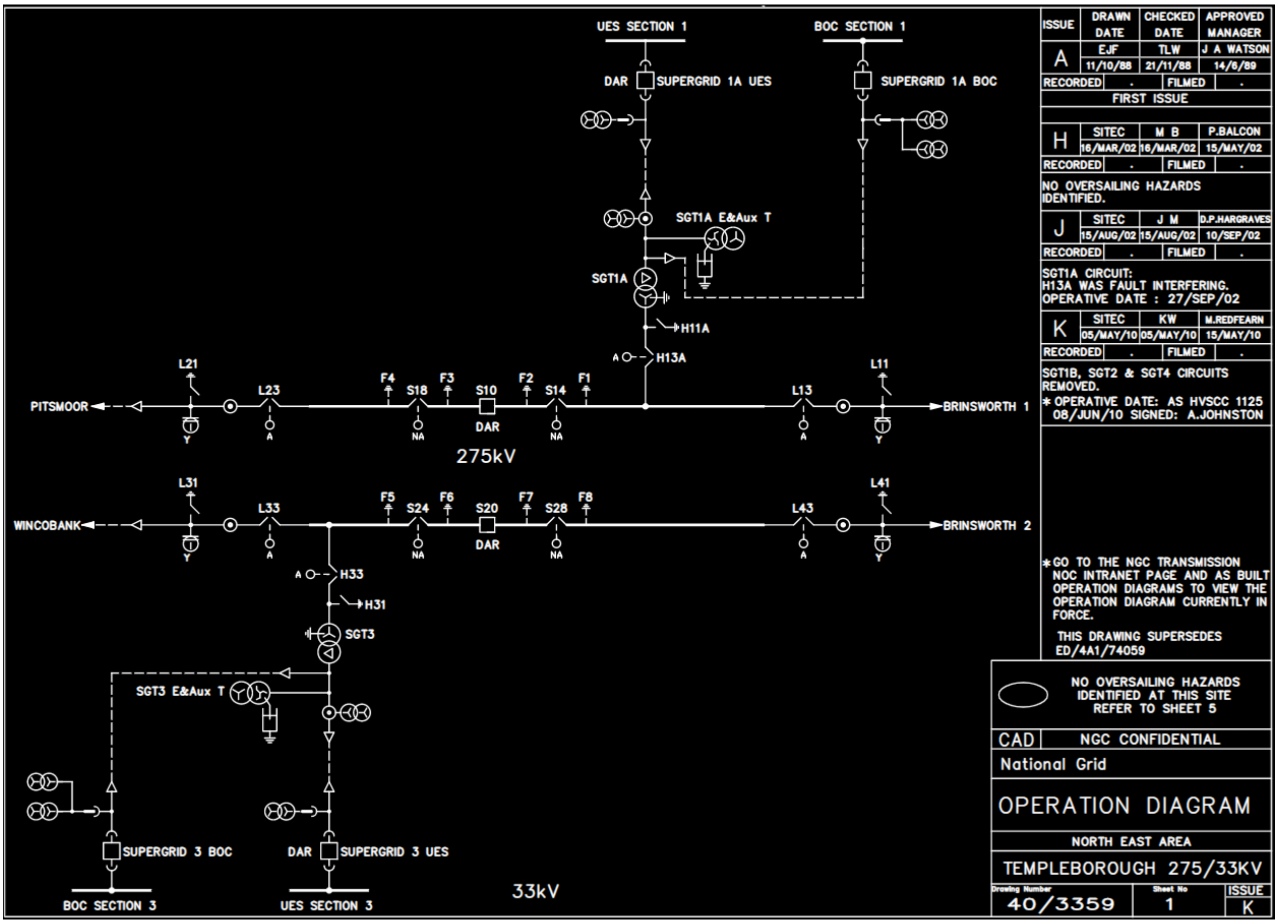
Alt: An alternative connection point flagged was MC2, but this would need to be confirmed based on the future retention of the Pitsmoor circuit.

Review of HV Bay Connections

Google Image:



Running Arrangement diagram



Indicative Completion Date for Connection

Scope of Work	Reconfigure Mesh Corner 4 and remove the old transformer plinth, bund and associated civil structures to create the space for a NGET owned connection bay.
Completion of NGET works (Indicative)	31st Oct 2031.
Indicative Connection / EISD (Indicative)	31st Oct 2031.

Note: The working assumption is the application for the connection would be made in / around ~ June 2026 or the next available application window to achieve the EISD.

Review on Outage Availability and Resource

No outages or resources have been secured at this time due to the uncertainty of the connection program.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	AIS Mesh modifications (£5,000k)
Connection costs (if applicable) *	Mesh ‘Connection Bay’ (£5,500k)
One-off costs (if applicable)	N/A
Demolition Costs	£1,500k
Total Cost (Without optional Connection Costs). **As this is a mesh corner substation connection this option is not available.**	**£12,050k**
Total Cost (With optional Connection Costs) *	**£12,050k**

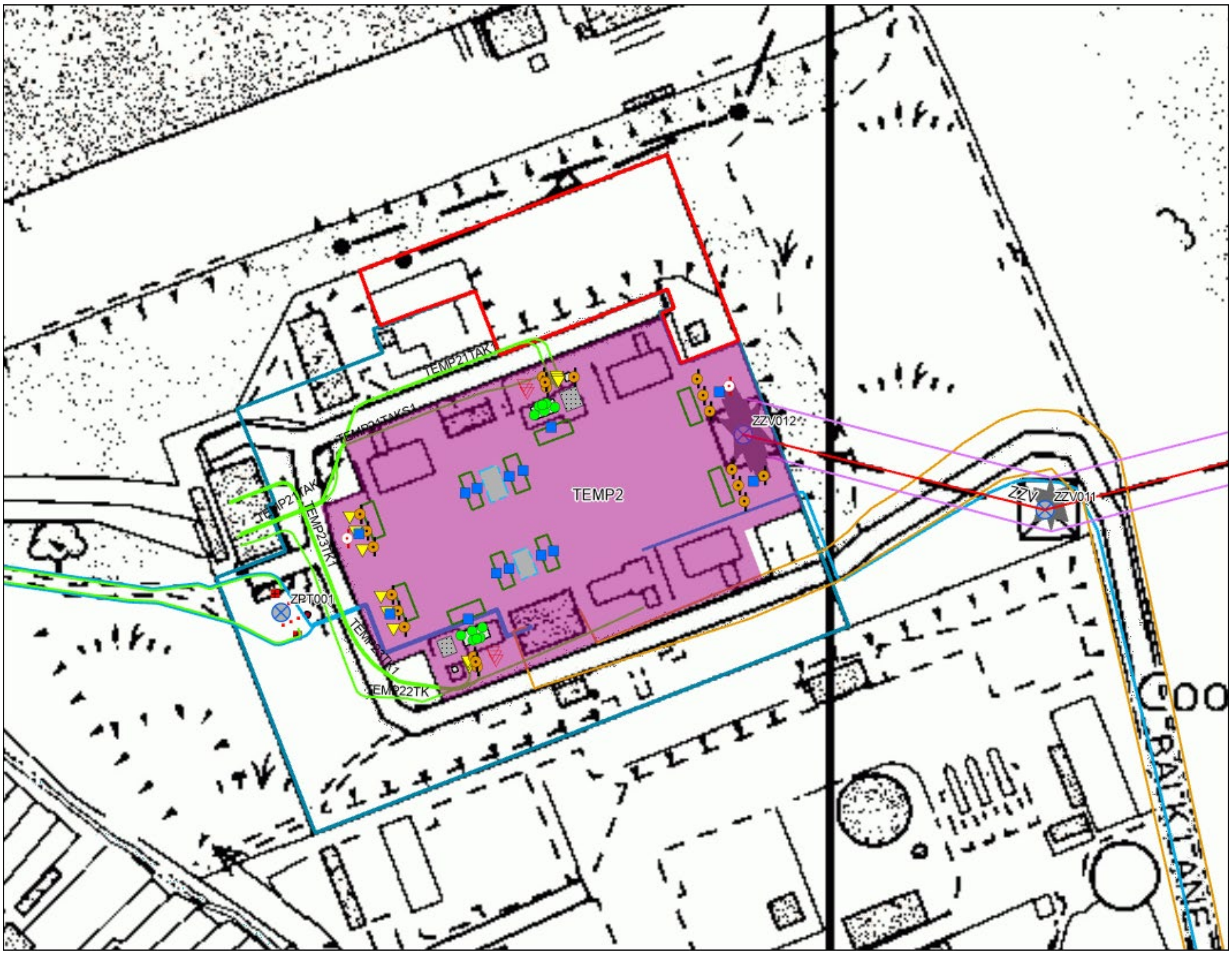
Note 1: These costs have indicative demolition costs for the plinth, bund and associated civil structures to create the space for a NGET owned connection bay. An indicative cost for these demolition and structure removal costs would be circa £1.5m. This £1.5M is in addition to the infrastructure costs at £10.5M and the application cost.

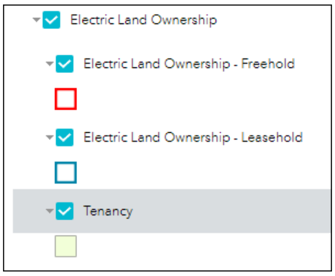
Note 2: Any impact on costs requiring either ARS or PoW are as follows:

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

Land Ownership:





Site Info	Response
Technical Limitation (TL's)	Yes. Not in the area that will create the PoC.
Risk Management Hazard Zones (RMHZ)	Yes
Operational Engineering Safety Bulletins (OESB)	Yes
ISS classification / security	Standard, no enhancements.
Flood Risk	No
Additional Site Information	Accessibility to the substation is via a winding road. The substation is located close to a BOC plant; there is minimal risk to NGET from those processing activities.

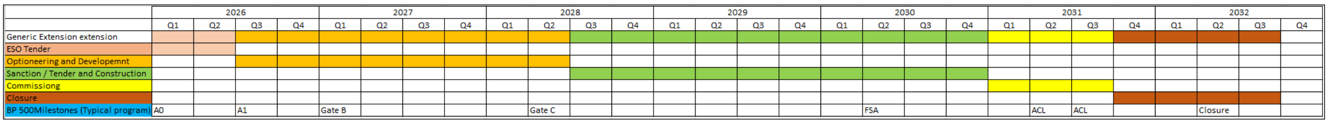
Load/Connection Offer Interactions

None

Non-Load Interactions

011424 - PITS-WIBA-TEMP Cable Replacement, ACL 30/09/2026

High Level Delivery Programme for Templeborough 275kV



Summary & recommendation

Reserved bay will be based on the reconfiguration of Mesh Corner 4 and removal of the old transformer plinth, bund and associated civil structures to create the space for a NGET owned connection bay for the service provider.

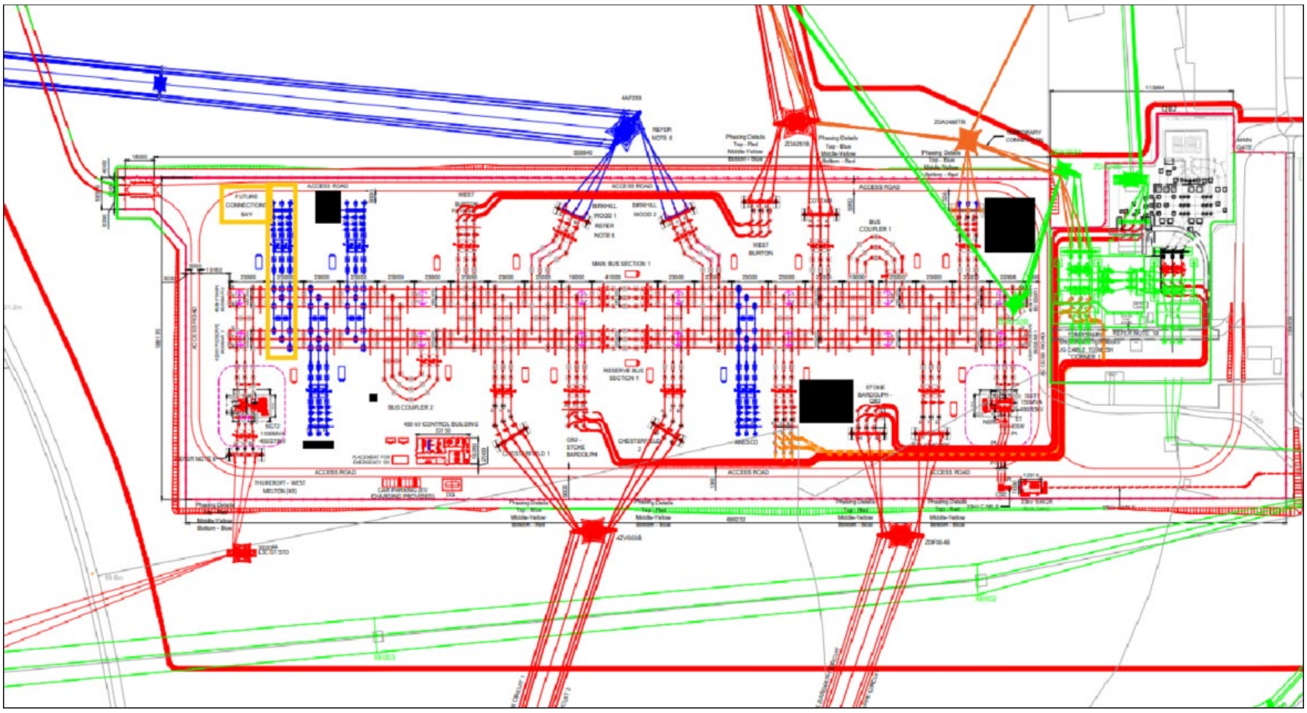
New High Marnham 400kV

Reserved Bay Request

High Marnham is having a new 400kV substation built directly adjacent to the current 400kV site. This is due to be constructed by ASTI with an anticipated completion date of 2029/30. This site will have space for an additional bay but would likely be restricted by the loss of infeed for generation connections already being close too or at capacity on both sections.

However, for 0 MW connections there is the possibility circa 2030/31.

The old High Marnham 400kV & 275kV substations are due to be demolished around the same time, as such the old 275kV site could be used for future expansion with a 3rd section post project completion (circa 2035).



Orange outline identifies the LT29 Bay (layout and PoC's need TBC during development).

Review of HV Bay Connections

Not available at this time.

Google Image:

Running Arrangement diagram

Not available at this time.

Indicative Completion Date for Connection

Scope of Work	Double busbar extension for a USER built connection bay.
Completion of NGET works (Indicative)	31st October 2029.
Indicative Connection / EISD (Indicative)	31st October 2031.

Note: The working assumption is the application for the connection would be made in / around June 2026 or the next available application window to achieve the EISD.

Review on Outage Availability and Resource

No outages or resources have been secured at this time due to the uncertainty of the connection program.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	£4,000k
Connection costs (if applicable) *	N/A
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£4,050k**
Total Cost (With optional Connection Costs) *	**£4,050k**

Note 1: This excludes the customer build bay. Users should account for their own costs.

Note 2: Any impact on costs requiring either ARS or PoW are as follows:

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

This will be a new outdoor 400kV AIS substation.

It will be constructed on land specifically procured for this purpose.

The operational and non-operational land boundaries are not yet confirmed.

Site Info	Response
Technical Limitation (TL's)	No (new build)
Risk Management Hazard Zones (RMHZ)	No (new build)
Operational Engineering Safety Bulletins (OESB)	No (new build)
ISS classification / security	The new High Marnham 400kV substation has been classified as requiring ISS classification and will require cyber secure access.
Flood Risk	The site is in floor risk zone 1, which is lowest risk. There is however a surface water risk NGET will need to mitigate during the design.
Additional Site Information	New build substation, as such the final design and layout are not finalised. Any relevant information will be communicated at the appropriate optioneering and development phase of any contract progression.

Load/Connection Offer Interactions

101874 - New High Marnham 400kV ss Anesco, ACL 31/10/2031

104220 - HIGB4 One Earth PSR 740MW PV-BESS, ACL 31/10/2029

104218 - HIGB4 ENSO GREEN HOLDINGS K 57MW, ACL 31/10/2029

102048 - Extend HIGM 400 SS for EGH user bay, ACL 31/10/2034

103357 - HIGM4 JCPGC 500MW BESS, ACL 30/07/2039

103376 - HIGM4 Morton Manor 240MW PV&BESS, 30/07/2039

102152 - HIGB4 Mosedale Energy BESS, ACL 31/10/2033

103759 - CHTE SOLARTEN 710MW PV/BESS, ACL 30/10/2034

102227 - HIGB4 JG Pears CHP Power Station GEN 2, ACL 31/10/2033

Non-Load Interactions

100459 - Uprating High Marnham - Stoke Bardolph, ACL 31/08/2026

100818 - CGNC - North Humber to High Marnham, ACL 28/07/2031

100907 - High Marnham - Thurcroft - West Melton, ACL 30/10/2026

101480 - Uprate High Marnham – Thurcroft – West Melton (West Melton Leg), ACL 31/10/2027

101481 - Uprate High Marnham – Thurcroft – West Melton (Thurcroft Leg), ACL 31/10/2027

101553 - Turn-in the Cottam-Staythorpe 400kV circuit at High Marnham 400kV, ACL 31/10/2031

101812 - EDEU - Brinsworth to High Marnham, ACL 26/07/2029

101873 - New High Marnham 400kV substation, ACL 29/10/2029

104348 - ZDA DISC HIGH MARNHAM ROUTE 248248, ACL TBC

101612 - HIGM: Second Interbus Transformer, ACL 30/10/2029

101714 - HIGM4 2x200 MVAr MSC's at new DBB Sub, ACL 31/10/2031

High Level Delivery Programme for New High Marnham 400kV Pathfinder connection

	2026				2027				2028				2029				2030				2031				2032			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																												
ESO Tender																												
Optioneering and Developemnt																												
Sanction / Tender and Construction																												
Commissiong																												
Closure																												
BP 500Milestones (Typical program)	A0		A1		Gate B					Gate C								FSA				ACL	ACL			Closure		

Summary & recommendation

The new 400kV AIS substation will have space for a reserved bay to enable the service providers Point of Connection. A reserved bay connection will be available circa 2031.

The exact bay allocated for the service provider will be determined as part of the new substation optioneering and design process.

Additional discussions will take place with the service provider as part of this design process to factor in the service providers requirements and an agreement will be reach between all parties to confirm the bay allocation for the service provider's Point of Connection.

Ocker Hill 275kV

Reserved Bay Request

Ocker Hill is a 275kV mesh corner substation. The land boundary outside of the substation land is likely to have protected wildlife and underground services which will need further investigation. The mesh corner will also need a reconfiguration to support the addition of a mesh corner customer bay. In recent years this site has been an area of interest but due to its location access to the site for schemes has been difficult.

Review of HV Bay Connections

Ocker Hill 275kV currently has one customer contracted to 2027, due to the sites land constraints a number of options have been explored with multiple aims:

1. Whats the most efficient to conenct the contracted customer
2. Can the site be future proofed
3. Which is the most cost efficient
4. Which option requires the least amount of work and outages
5. MSC 4 is the only feesable connection point

Currently there is only one currently contracted BESS acceleration connection, with an ACL of 2027, due to site location and works included for future extension this is the only customer concidered.

For the extra bay reservation to be deemed feasible, NGET would need confirmation and security of what is being proposed at the site. The current, most favoured option (NGET Option 1 C) by all within NGET currently would only support the need for one contracted connection (currently contracted) as per contracted requirements. This would mean no reserved bay being available for an LT29 Scheme. There is an option (NGET Option 4) which would enable the connection of the current contracted BESS and the LT29 scheme however significant delays and alteration of current easements would need to take place. The difference in enabling works can be seen below in Figure Ocker Hill 275kV Options.

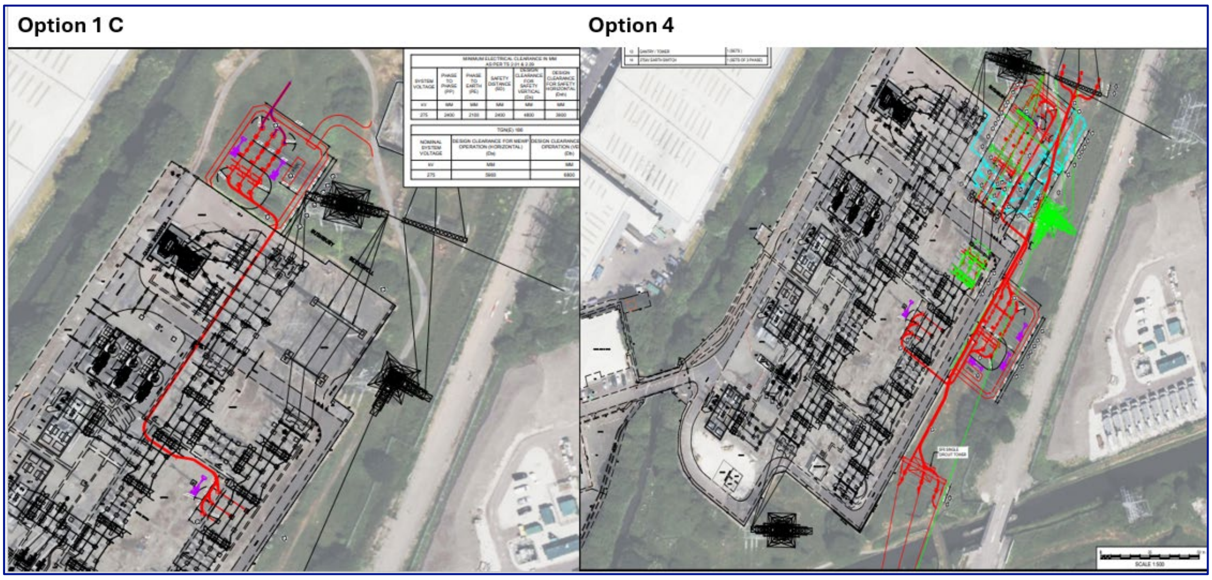
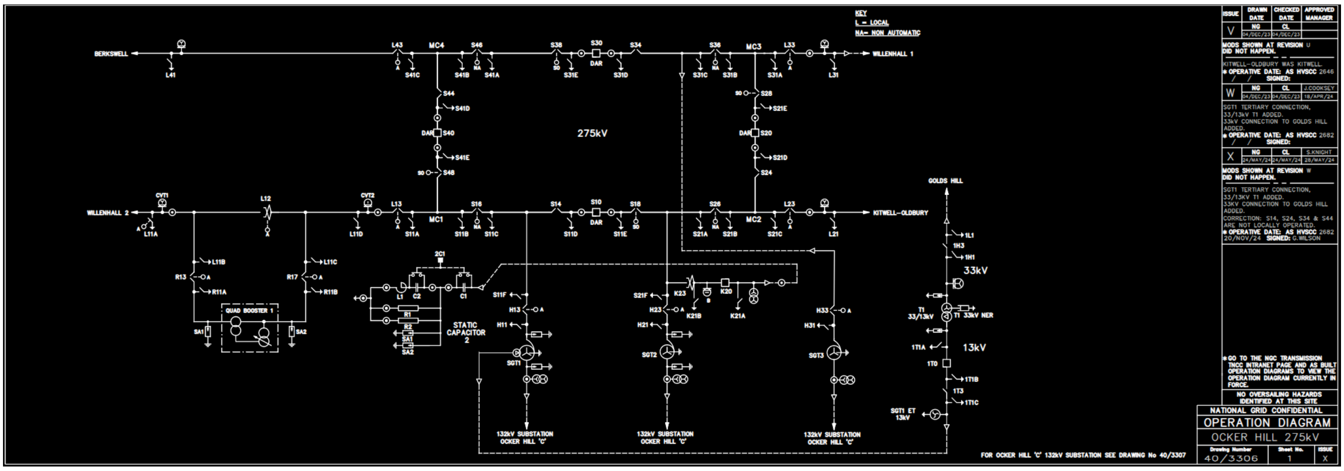


Figure Ocker Hill 275kV NGET Options

Google Image:

Running Arrangement diagram



Indicative Completion Date for Connection

Scope of Work	Please refer to option 4: To facilitate extra connections at the site substantial work would need to take place and reconfiguration of the site. This work would include an extension of the bars, cabling work,
Completion of NGET works (Indicative)	31st October 2032.
Indicative Connection / EISD (Indicative)	31st October 2032.

Review on Outage Availability and Resource

No outages or resources have been secured at this time due to the uncertainty of the connection program.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	AIS Mesh modifications (5,000k) Tower diversion and OHL works (£1,500k) Cabling (£300k)
Connection costs (if applicable) *	Mesh 'Connection Bay' (5,500k)
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs). **As this is a mesh corner substation connection this option is not available.**	**£12,350k**
Total Cost (With optional Connection Costs) *	**£12,350k**

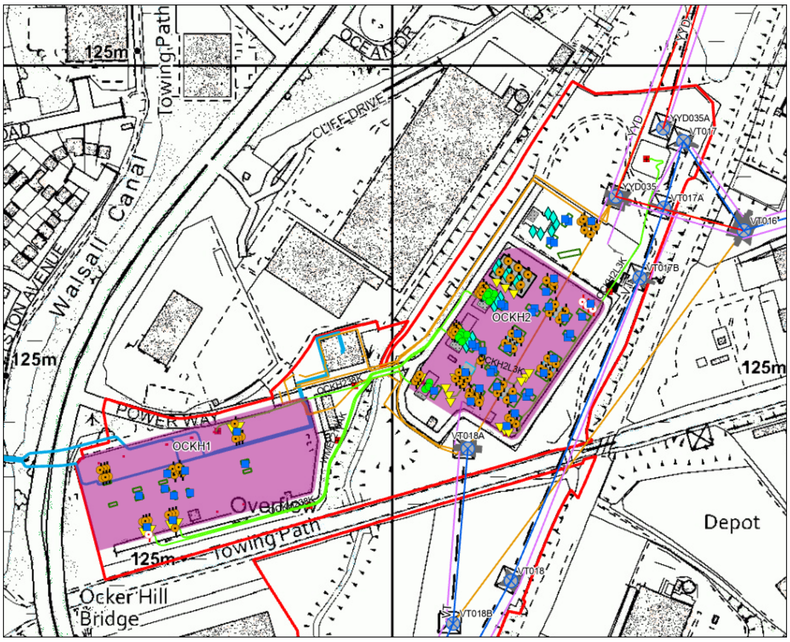
Note 1: Any impact on costs requiring either ARS or PoW are as follows:

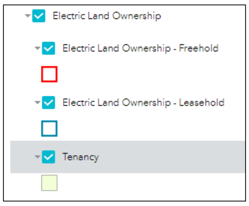
Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

Substation located within the outer part of Birmingham. The operational boundary is marked in the picture below; the non-operational boundary is also marked within the red site boundary. Ocker Hill is suspected to harbour some protected wildlife highlighted in early ecological reports. The main bulk of works associated with the proposed new bay, or any subsequent works would be ground works and cable routing, cabling into the site is also known to be a reoccurring issue due to its location and surrounding land.

Land Ownership:





Site Info	Response
Technical Limitation (TL's)	Yes
Risk Management Hazard Zones (RMHZ)	No
Operational Engineering Safety Bulletins (OESB)	Yes
ISS classification / security	Standard, no enhancements.
Flood Risk	Yes
Additional Site Information	Access is constrained on this site; other third parties such as network rail and other landowners are present.

Load/Connection Offer Interactions

102113- Murphy Distribution Centre Ocker Hill, contracted ACL October 2027

Non-Load Interactions

104342- YYD OCKER HILL – WILLENHALL

103679 - HAMH 275kV Cable Replace (BESW – OCKH)

104352 - ZF BERKS-OCKER-DRAKE-OLDB

High Level Delivery Programme for Ocker Hill 275kV

	2026				2027				2028				2029				2030				2031				2032			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																												
ESO Tender																												
Optioneering and Developemnt																												
Sanction / Tender and Construction																												
Commissiong																												
Closure																												
BP 500Milestones (Typical program)	A0		A1		Gate B					Gate C								FSA				ACL	ACL			Closure		

Summary & recommendation

The conclusion on Ocker Hill is that to support 1 x extra bay at the site for a reserved bay, NGET would need a decision as to if the LT29 scheme is deemed feasible for the site and support will be needed to manage the affected BESS customer. The works involved would firstly be expensive and extensive to support the two connections, therefore this site is not a recommended location by NGET for LT29 schemes.

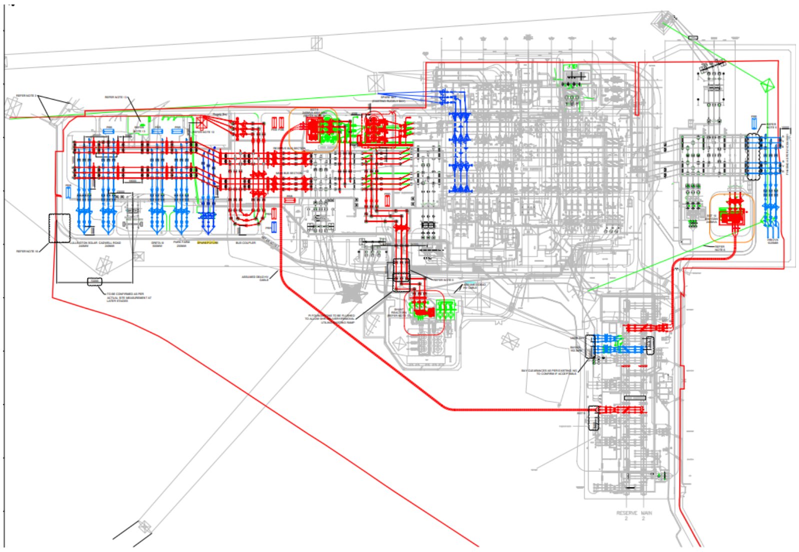
Drakelow 400kV

Reserved Bay Request

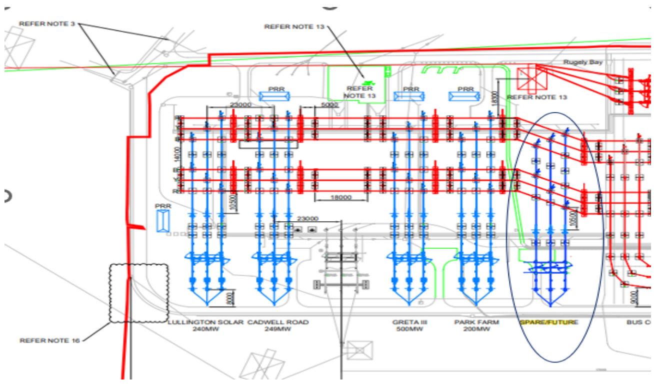
Drakelow has had a significant amount of interest when it comes to connections over the years, due to the amount of interest in the site and potential for connections in Derbyshire the approach has been taken to optioneered the site to its full potential with other NGET interactions to be considered. This site has previously been identified as a strategic site for connections.

Review of HV Bay Connections

Below is a screen shot showing the overall proposed site strategy of Drakelow 132kV and 400kV sites. In this site strategy the aim was to develop the site to its full capability taking into account the parcel of land owned by NGET and other schemes within the area and site. As part of this optioneering process it was identified that a there may be up to two bays available for future connections, the most feasible has been circled in figure Drakelow 400kV Connections. There are no provisions for spare bays at the 132kV site.



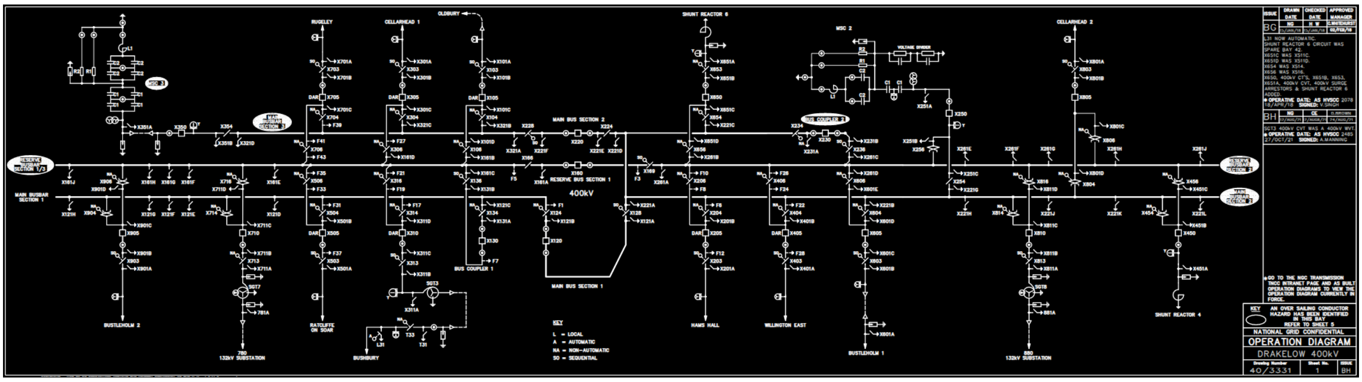
Drakelow Site Strategy Overview



Drakelow 400kV Connections

Google Image:

Running Arrangement diagram



Indicative Completion Date for Connection

Scope of Work	Busbar Extensions and bay construction
Completion of NGET works (Indicative)	31st October 2031.
Indicative Connection / EISD (Indicative)	31st October 2031.

Review on Outage Availability and Resource

No outages or resources have been secured at this time due to the uncertainty of the connection program.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	AIS Busbar Extension (£4,000k)
Connection costs (if applicable) *	N/A
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£4,050k**
Total Cost (With optional Connection Costs) *	**£4,050k**

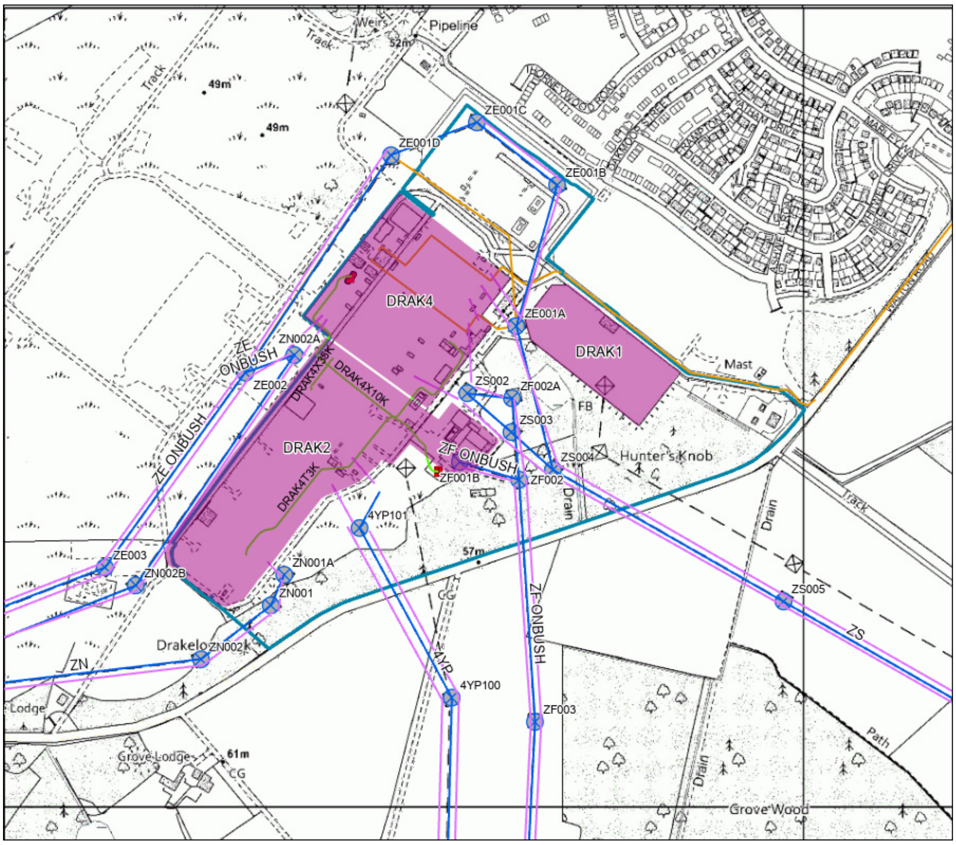
Note 1: Any impact on costs requiring either ARS or PoW are as follows:

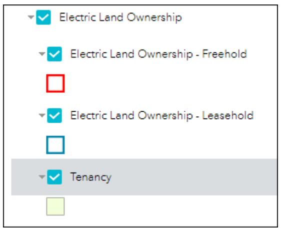
Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

This site is known to be a difficult site to cable into and gain planning permission. This is due to where its located and number of pre-existing cables in the ground and around the old PowerStation land. There is also a new large housing development being constructed close to the site and the proposed extension.

Land Ownership:





Site Info	Response
Technical Limitation (TL's)	Yes
Risk Management Hazard Zones (RMHZ)	Yes
Operational Engineering Safety Bulletins (OESB)	Yes
ISS classification / security	Standard, no enhancements.
Flood Risk	No
Additional Site Information	This site will be a busy site over the coming years with multiple NGET schemes working in the area and site. Planning is known to be a major constraint at this site.

Load/Connection Offer Interactions

100867-Baywa Connection

100582- Extension of the existing Drakelow 400kV substation Main and Reserve

Busbars to accommodate the User's Bay.

102085- Addition of two new section breakers and a bus coupler

102086- Circuit Transfer from existing substation to newly extended part of

Substation.

102707- Extension of the existing Drakelow 400kV substation Main and Reserve Busbars to accommodate the User's Bay.

102485 -Install SGT hot-standby scheme at Drakelow substation.

101615 - Greta III - Drakelow Customer Connection

101882 - 200MW BESS-Drakelow 400kV s/s Harmony EL

102067 - Drakelow 400kV-Bay ext. - Lullington

102909 - DRAK4 - IB VOGT UK LTD

Non-Load Interactions

032021 - Cellarhead-Drakelow overhead line reconductor

033207 - 4YP BUST - DRAK 1&2 (1 - 101) OHL Refurb

101891 - Drak4-Wile4 Uprating

101905 - RUGE-DRAK cct 8ohm Power control device

101907 - DRAK-HAMH reconductoring for 2490 MVA

101939 - Drakelow - Rugeley Recon ZN Route

101943 - Drakelow - Ratcliffe Hotwire - ZS-ZL-ZLA

102067 - Drakelow 400kV-Bay ext. - Lullington

102115 - DRAK4: WPD PP - SGT Uprating

102382 - CDP123 - PFC Cellarhead Drakelow

102907 - New section at Drakelow 400kV and 2CBs

102908 - Two circuit transfer at Drakelow

102916 - Drakelow-Bushbury Turn in Bushbury B 400

102917 - Drakelow-Bushbury B cct uprating

102918 - Relocate Interbus SGT Drakelow to Bushbury.

103028 - NGED WM Drakelow-Old Cct Turn in

103085 - Rats - Drake Turn in at Ratcliffe B 400

104687 - T3 - Reactive Comp - DRAK2

High Level Delivery Programme for Drakelow 400kV

	2026				2027				2028				2029				2030				2031				2032			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																												
ESO Tender																												
Optioneering and Developemnt																												
Sanction / Tender and Construction																												
Commissiong																												
Closure																												
BP 500Milestones (Typical program)	A0		A1		Gate B					Gate C								FSA				ACL	ACL			Closure		

Summary & recommendation

This 400kV site is a possible location for LT29 schemes however, it’s worth noting that access to the site is limited and constrained as previously mentioned. It’s of the projects team’s opinion that LT29 schemes can be facilitated at this site around 2031.

New Kemsley 400kV

Reserved Bay Request

The existing Kemsley 400 kV site is full with planned expansion works for other customers taking place. A second 400kV substation has been triggered with a siting study currently taking place with final location of the new substation to be determined Spring 2026.

The design of the substation will be confirmed and allocated as part of the new substation optioneering and design phases.

Review of HV Bay Connections

A bay will be allocated to the customer as part of the works to build the new substation.

Running Arrangement diagram

TBC. It will be a standard double busbar substation.

Indicative Completion Date for Connection

Scope of Work	Construction of a new 400kV AIS/GIS substation to facilitate the connection of multiple customers including the LT29 scheme onto the transmission network.
Completion of NGET works (Indicative)	31st October 2033.
Indicative Connection / EISD (Indicative)	31st October 2033.

Review on Outage Availability and Resource

Resources and outage confirmation will be reviewed as the scheme progresses to ensure the LT29 requirements are catered for as part of the optioneering / development and delivery phases of the site delivery progression.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	GIS Substation & Busbar Extension (5,500k)
Connection costs (if applicable) *	GIS Bay (Connection Asset) (5,500k)
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£5,550k**
Total Cost (With optional Connection Costs) *	**£11,050k**

Note 1: the new substation is anticipated to be in the range £90-£130m (dependant of technology type and location). This new infrastructure work will need to be completed to allow a LT29 connection.

Note 2 Any impact on costs requiring either ARS or PoW are as follows:

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

Site location unknown now pending outcome of siting study. View is to site it as close to Kemsley substation.

Load/Connection Offer Interactions

N/A currently, but subject to change.

Non-Load Interactions

N/A currently, but subject to change.

High Level Delivery Programme for New Kemsley 400kV

Indicative

	2026				2027				2028				2029				2030				2031				2032			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																												
ESO Tender																												
Optioneering and Developemnt																												
Sanction / Tender and Construction																												
Commissiong																												
Closure																												
BP 500Milestones (Typical program)	A0		A1		Gate B					Gate C								FSA				ACL	ACL			Closure		

Planned

ACL – Q3 2032

*Subject to substation siting study to confirm final location of the substation and a Town and Country Planning application.

Summary & recommendation

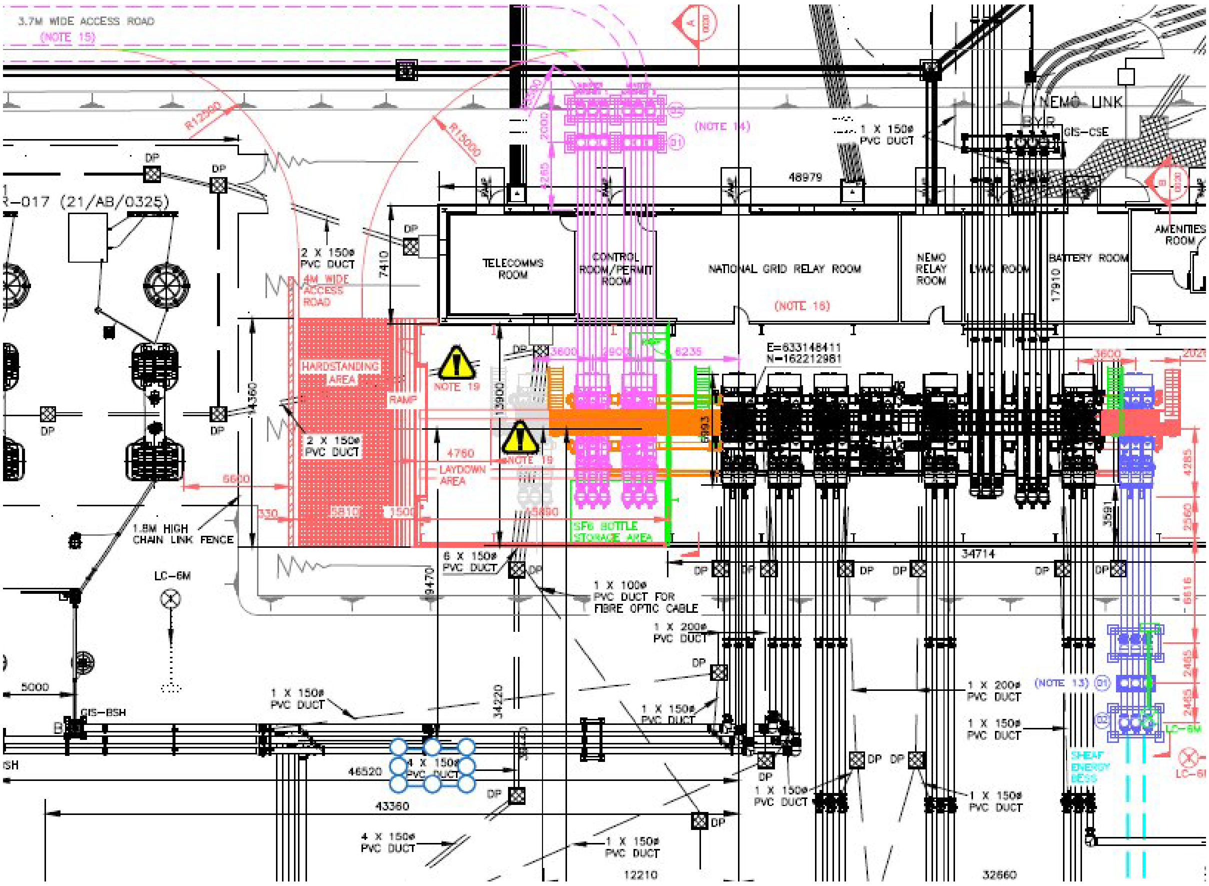
North Kent Connection Node A is a new substation that will be built close to the existing Kemsley substation. Final location TBC pending siting study with an indicative preferred location to be identified in spring 2026. The new substation will reserve a spare bay for an LT29 scheme.

Richborough 400kV

Reserved Bay Request

Note: Richborough reserved bays (2 bays) are the result of a customer connection terminating at that site. As such some optioneering and development work already exists.

Provide a building extension to the west of the existing GIS hall to provide space for a customer connection.

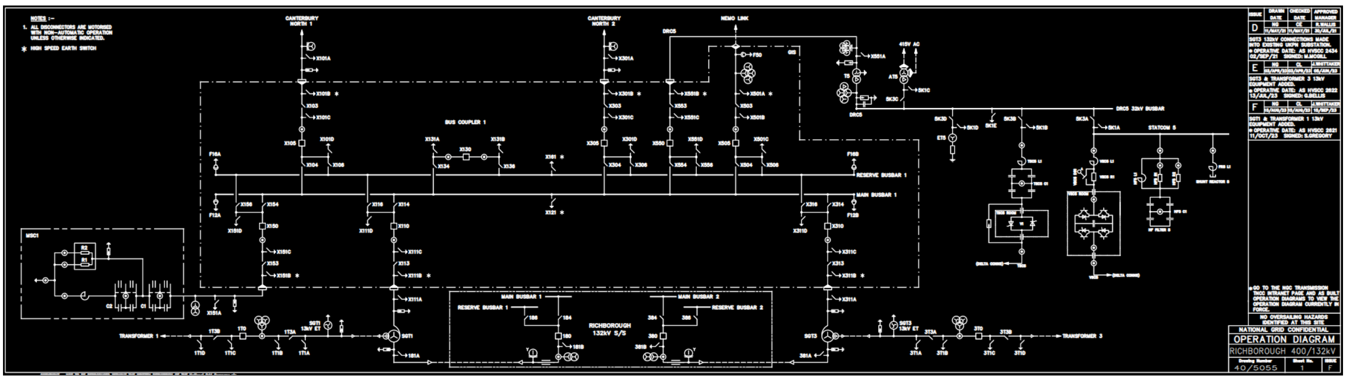


Review of HV Bay Connections

The remaining spare bay within the existing GIS hall to the east end has been allocated to a customer. Based on previosuly developed scope at Richborough, there is space for two additioanl customer bays to the west of the substation as indicated above. The GIS hall will be extended to accommodate the additional bays which will be allocated to the LT29 scheme(s).

The substation is surrounded by third party land that NGET has no control over. This means that negotiating a cable route into our substation will be a challenge as the developer has earmarked most of the land surrounding the substation for future development. The most likely scenario to bring a cable into the substation will be via horizontal directional drilling.

Running Arrangement diagram



Indicative Completion Date for Connection

Scope of Work	• Richborough 400kV GIS building extension to the west for two User's bays. • New Access Road / hard standing / laydown area • Protection and control as required (busbar protection/ SCS database update) • NGET Gas zone to enable bay connection is expected to be provided by the User due to nature of compact GIS as part of GIS customer bay order. OEM's have declined to offer busbar gas zone on its own. • New compliant Amenities Facility
Completion of NGET works (Indicative)	31^{st} October 2029.
Indicative Connection / EISD (Indicative)	31^{st} October 2029.

Review on Outage Availability and Resource

Busbar outages for bay busbar extension deemed feasible.

SAP resources in the southeast region limited. Request for resources to deliver the works will be submitted.

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	GIS Substation & Busbar Extension (£5,500k)
Connection costs (if applicable) *	GIS Bay (Connection Asset) (5,500k)
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£5,550k**
Total Cost (With optional Connection Costs) *	**£11,050k**

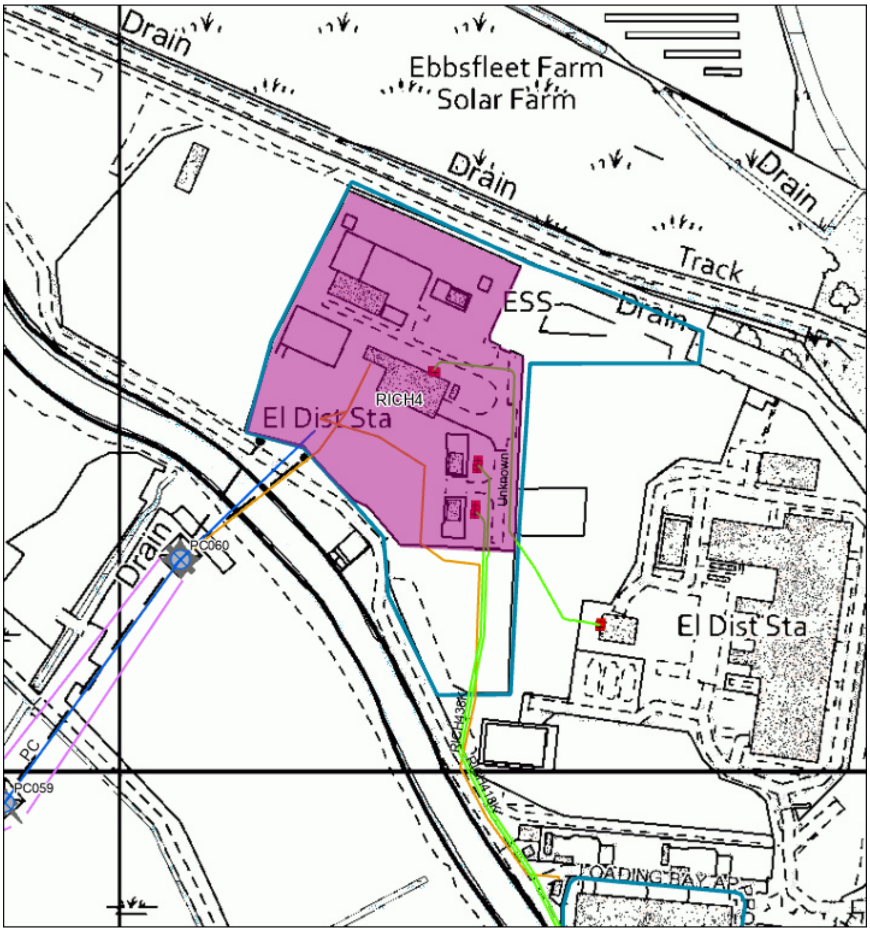
Note 1: Any impact on costs requiring either ARS or PoW are as follows:

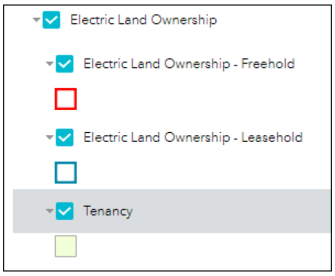
Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

Land Ownership:

The Richborough substation is a long leasehold and the land around the substation is owned by a third party. Any cable easements to enter our substation will have to negotiated via third party and not NGET. Previous discussions with the landowner around Richborough suggest that this won't be a straightforward exercise and horizontal directional drilling (HDD) may be the only viable option for a cable route.





Site Info	Response
Technical Limitation (TL's)	No
Risk Management Hazard Zones (RMHZ)	No
Operational Engineering Safety Bulletins (OESB)	No
ISS classification / security	Standard Security measures at present.
Flood Risk	No
Additional Site Information	Shared access road with 3rd party landowner.

Load/Connection Offer Interactions

102345 - 102345 Sheaf Energy RICH400 249MW BESS.

Non-Load Interactions

103739 - TWNC 400kV Richborough2 - Aldington

High Level Delivery Programme for Richborough 400kV

	2026				2027				2028				2029				2030				2031				2032			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																												
ESO Tender																												
Optioneering and Developemnt																												
Sanction / Tender and Construction																												
Commissiong																												
Closure																												
BP 500Milestones (Typical program)	A0		A1		Gate B					Gate C								FSA				ACL	ACL			Closure		

Summary & recommendation

There is space available within Richborough 400kV GIS substation to extend the GIS hall to the west for the LT29 bay. NGET will undertake the GIS hall extension works with the customer responsible for procuring and installing the GIS bay.

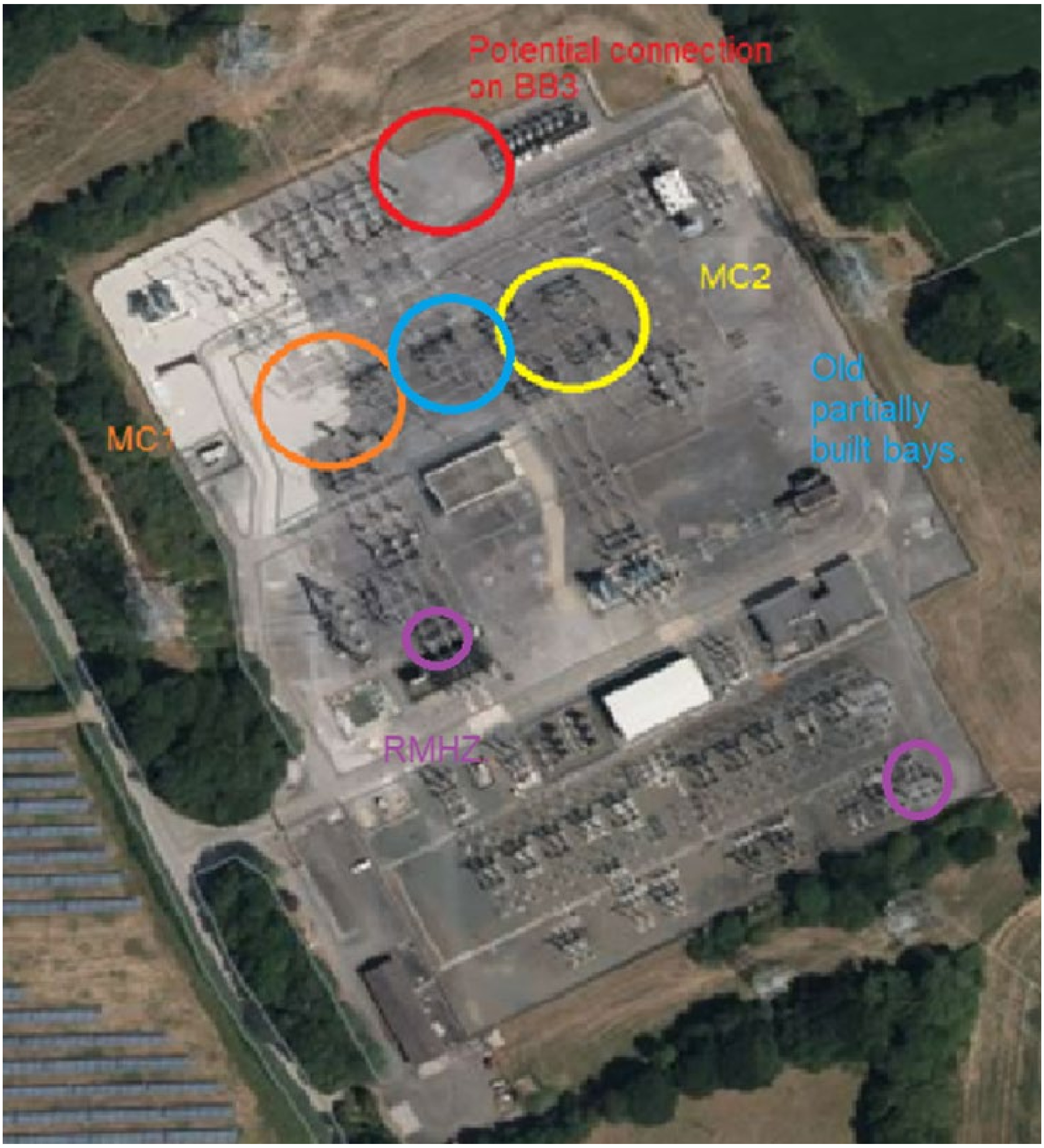
Landulph 400kV

Reserved Bay Request

The connection available at Landulph 400kV is for two individual bays is the result of a previous customer terminating at the site. As such, some optioneering and development work exists.

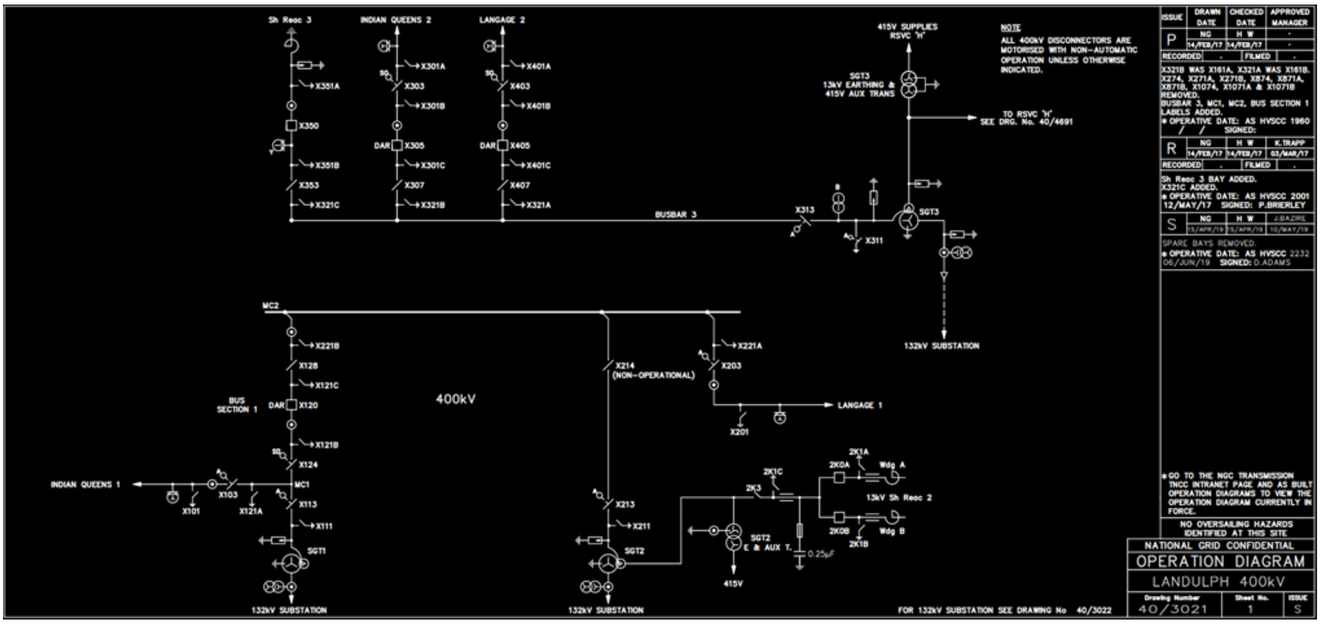
The proposed LT29 connection for two reserved bays will be to a single busbar, with the installation of a new CB at SGT3 terminal (Busbar 3) required in line with works. Some consideration in the planning phase will need to be given to OHL proximity and forthcoming maintenance works (Langage OHL route YF) and 13kV RSVC compound. RMHZ's will need to be considered throughout construction activities but are unlikely to pose any significant civils or M&E issues

No connections are available on MC1 & MC2 due to the stability of connection and space adjacent to existing plant. Included in image below for information purposes only.



Review of HV Bay Connections

Running Arrangement diagram



Indicative Completion Date for Connection

Scope of Work	• Landulph 400kV extension to incorporate single busbar and separate single switch mesh arrangement. • Protection and control as required. • Connection made to Busbar 3 adjacent to Langage circuit. • Customer to facilitate cabling to bay. • Connection boundary to be agreed. • External land rights and associated works to be facilitated by customer
Completion of NGET works (Indicative)	31st October 2030.
Indicative Connection / EISD (Indicative)	31st October 2030.

Review on Outage Availability and Resource

Bar outages and resources expected to be available during civils and M&E phases, however, these cannot be presently determined

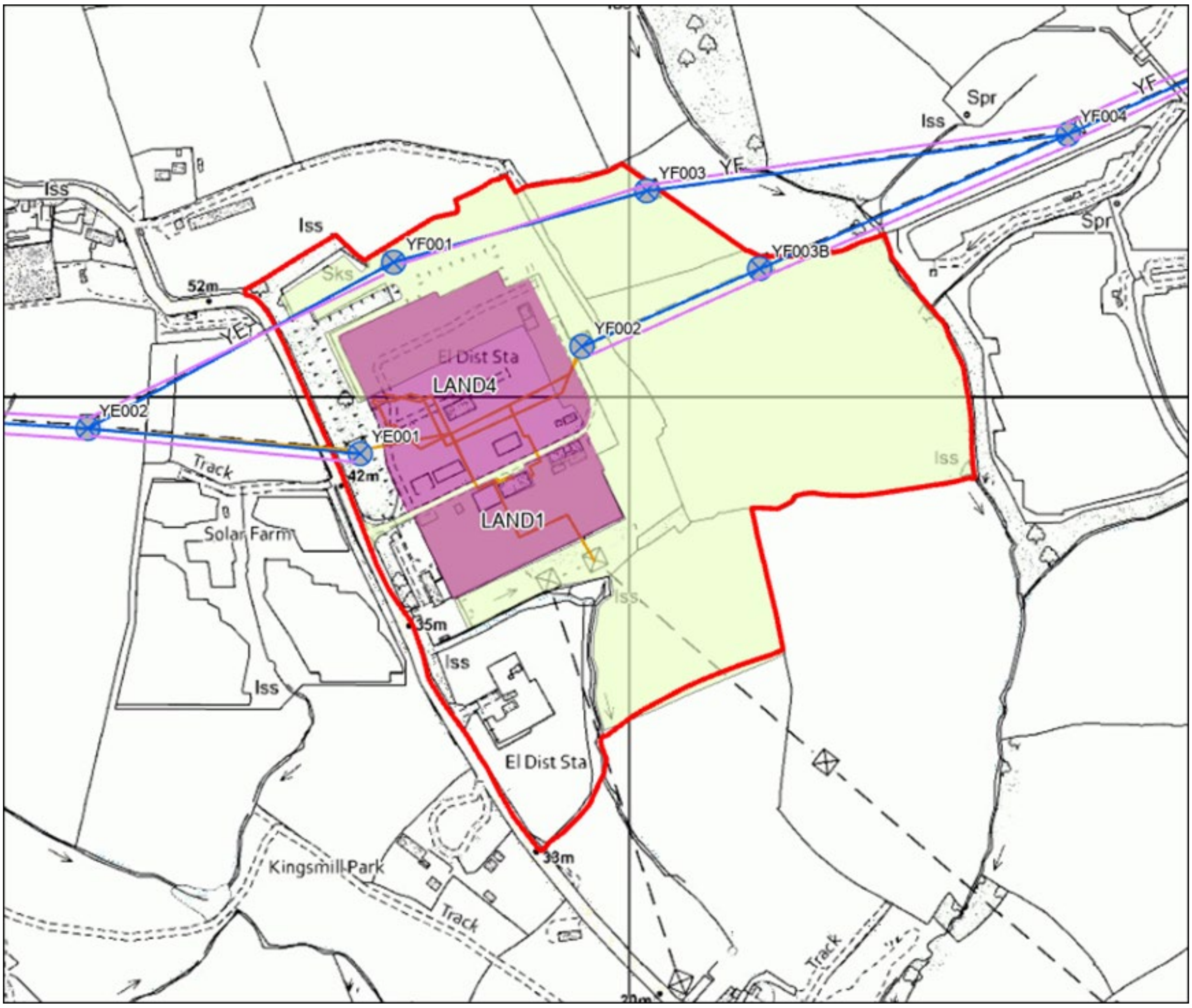
Indicative Cost Estimate for works

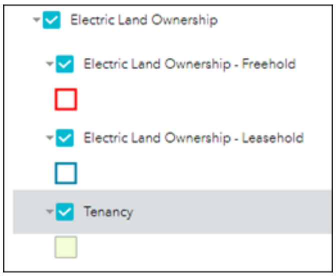
Application Cost	£50k
Infrastructure Costs	£4,000k
Connection costs (if applicable) *	N/A – user bay allocated
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£4,050k**
Total Cost (With optional Connection Costs) *	**£4,050k**

Note 1: Any impact on costs requiring either ARS or PoW are as follows:

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge





Site Info	Response
Technical Limitation (TL's)	Proximity of OHL (YF) & connection to busbar
Risk Management Hazard Zones (RMHZ)	2no. RMHZ's. Surge Arrestor works have been undertaken on SGT1 to alleviate RMHZ.
Operational Engineering Safety Bulletins (OESB)	None known
ISS classification / security	Standard S/S security arrangements
Flood Risk	Potential surface water flood risk – site under consideration for flood defence works
Additional Site Information	Civils works on a separate Tertiary project currently underway on substation

Load/Connection Offer Interactions

None identified at this time.

Non-Load Interactions

102449 - LAND4 SGT3 HV CB installation

102476 -LAND4 SGT3 CB for complexity rules

102477 - LAND4 MC2 HV CB installation

104413 - LAND4 2KY Replacement

High Level Delivery Programme for Landulph 400kV

Indicative:

	2026				2027				2028				2029				2030				2031				2032			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																												
ESO Tender																												
Optioneering and Developemnt																												
Sanction / Tender and Construction																												
Commissiong																												
Closure																												
BP 500Milestones (Typical program)	A0		A1		Gate B					Gate C								FSA				ACL	ACL			Closure		

Work has the potential to commence at Landulph 400kV circa 2028/29

Summary & recommendation

LT29 scheme will consist of a single busbar and separate single-switch mesh arrangement made at Busbar 3 adjacent to Langage circuit 2, this is to provide two separate Points of Connection.

Adjacent external land acquisitions, land works and cable easements would need to be determined by the customer.

New Woolavington 400kV

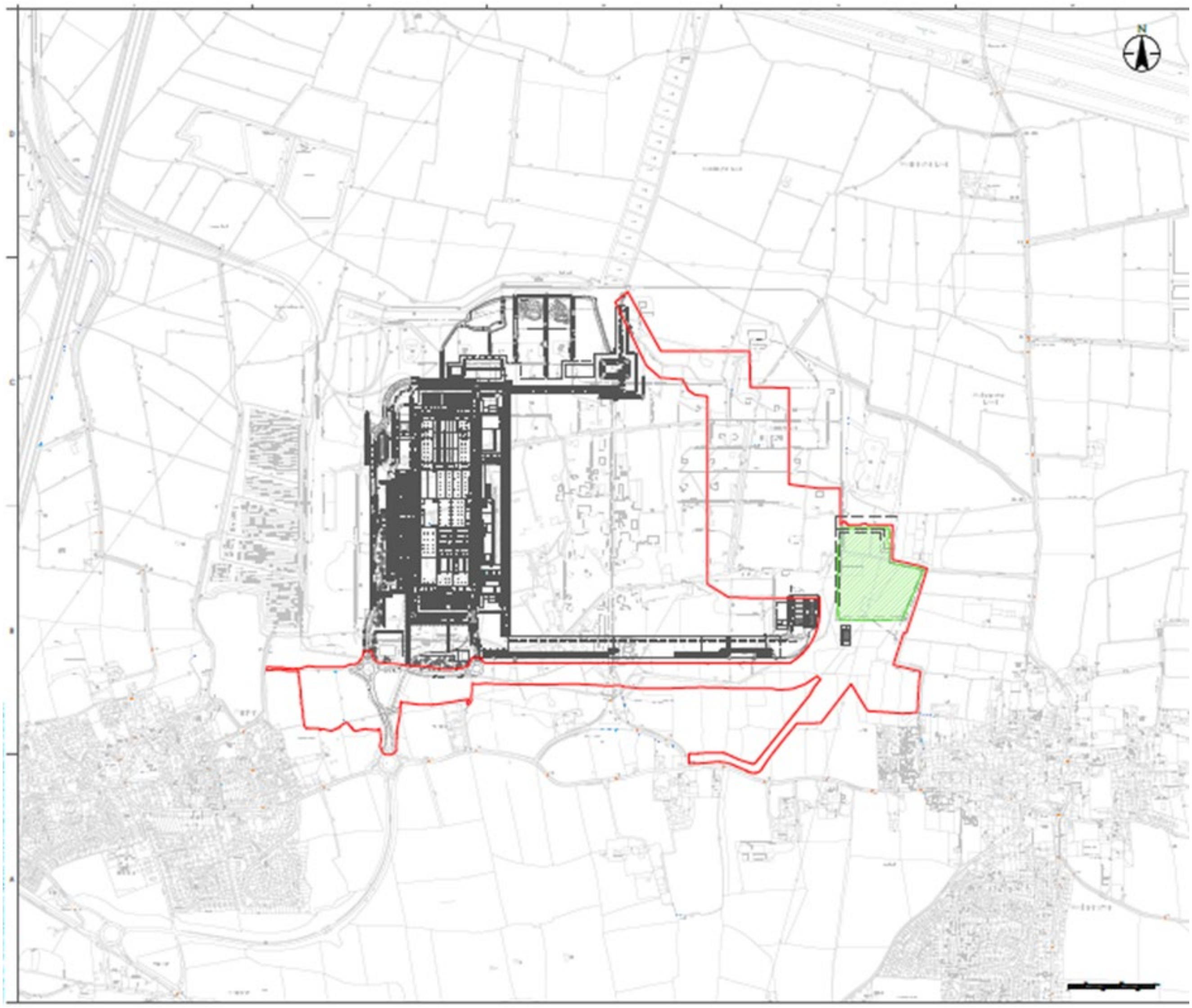
Reserved Bay Request

Woolavington 400kV is a new substation currently being constructed east of Bridgwater. It will be configured as a double bus substation using Gas Insulated Switchgear.

A bay has been reserved by NESO for LT29, identified likely as future feeder bay 5 to the far east of the 400kV GIS hall.

The connection will be a 400kV bay connection with associated switchgear.

It is anticipated that the connection will be for an ACL date of circa 31st October 2030.

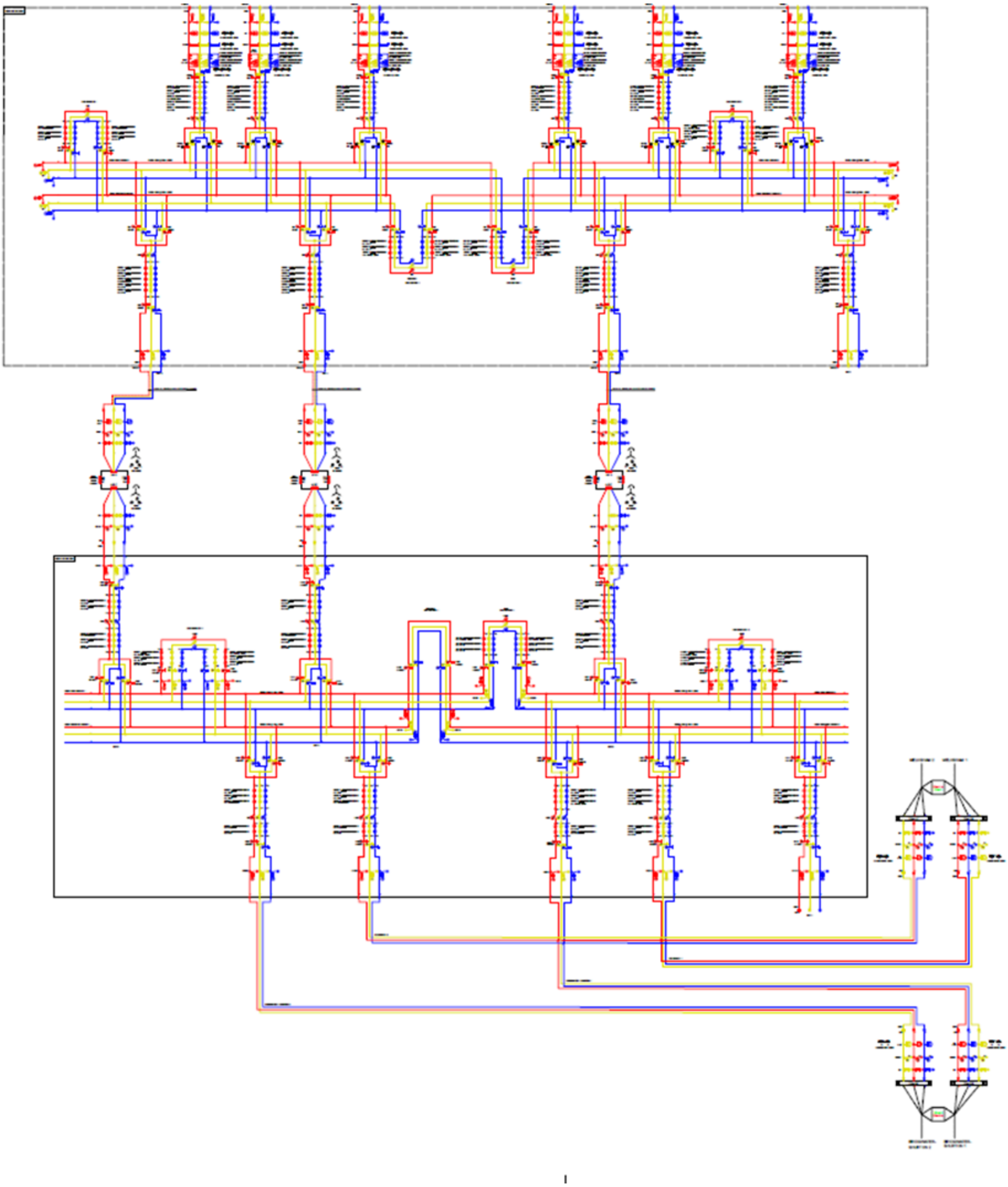


Indicative location of substation highlighted in green

Review of HV Bay Connections

The LT29 bay will be connected to NGET 400KV GIS Hall. NGET do not have land at Woolavington to offer customers. Any LT29 project would need to be located offsite of the substation, connected via cables to the feeder bay.

Running Arrangement diagram



Indicative Completion Date for Connection

Scope of Work	• Woolavington 400kV GIS building to the east for single bay connection with SF6-free T155-7 switchgear • Protection and control as required. • Customer to facilitate cabling to bay. • Connection boundary to be agreed
Completion of NGET works (Indicative)	31st October 2030.
Indicative Connection / EISD (Indicative)	31st October 2030.

Review on Outage Availability and Resource

Bar outages and resources expected to be available post completion of Woolavington S/S works

Indicative Cost Estimate for works

Application Cost	£50k
Infrastructure Costs	GIS Substation & Busbar Extension (£5,500k)
Connection costs (if applicable) *	GIS Bay (Connection Asset) (£5,500k)
One-off costs (if applicable)	N/A
Total Cost (Without optional Connection Costs)	**£5,550k**
Total Cost (With optional Connection Costs) *	**£11,050k**

Note 1: Any impact on costs requiring either ARS or PoW are as follows:

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

NGET Site Knowledge

The substation is surrounded by third party land that NGET has no control over and therefore cable easements over non-NGET land would have to be acquired via the landowner

Site Info	Response
Technical Limitation (TL's)	N/A new site
Risk Management Hazard Zones (RMHZ)	N/A new site
Operational Engineering Safety Bulletins (OESB)	N/A new site
ISS classification / security	TBC
Flood Risk	Flood mitigation to be installed
Additional Site Information	FSA October 2025 (main project)

Load/Connection Offer Interactions

Supplying 2no. customers at 132kV for a battery factory and data centre – ACL 04/2029

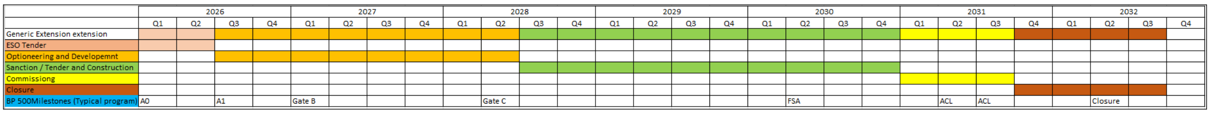
Supplying 5no. 460 MVA SGT's, 3no. to be installed in initial substation construction phase – ACL 04/2029

Non-Load Interactions

None identified at this time.

High Level Delivery Programme for New Woolavington 400kV

Indicative:



Work to commence on new Woolavington 400kV substation (Q3 2025).

Estimated ACL for pathfinder Q4 2030.

Summary

The new 400kV GIS substation will have space for additional bays to facilitate the reserved bay.

A reserved bay connection will be available circa 2030, but this is dependent on the new GIS substation build progressing. Otherwise, Woolavington 400kV has no place for any connection.

The exact bay allocated for the service provider will be determined as part of the new substation optioneering and design process.

NGET Investment Programme Summary of Reserved Bays

Build Indicative Program:

(Scheme kick off and ACL dates need to be adjusted to reflect the offered ACL date):

	2026				2027				2028				2029				2030				2031				2032			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																												
ESO Tender																												
Optioneering and Developemnt																												
Sanction / Tender and Construction																												
Commissiong																												
Closure																												
BP 500Milestones (Typical program)	A0		A1		Gate B					Gate C								FSA				ACL	ACL			Closure		

Asset Transfer Indicative Program:

(Scheme kick off and ACL dates need to be adjusted to reflect the offered ACL date):

	2026				2027				2028				2029				2030			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Generic Extension extension																				
ESO Tender																				
Optioneering and Developemnt																				
Sanction / Tender and Construction																				
Commissiong																				
Closure																				
BP 500Milestones (Typical program)	A0		A1		Gate B			Gate C			FSA				ACL		Closure			

Site	Scope of Works	Connection Option (£k)	Infrastructure cost (£k +/- 25%)	NGET TP500 Runway Ref	EISD #1	Scope Complexity Ranking	Note
Greystones A/B 275kV	GIS Bay transfer. 1. P&C. 2. Database (SCS). 3. Commissioning.	GIS Asset Transfer (£1)	£250k	N/A, we are using a special runway specific for LT29.	2030	Simple	Customer bay will be located at Substation A or B TBC. NGET are not offering connection assets.
Penwortham 400kV	GIS Substation & Busbar Extension 1. Extend the building. 2. Extend the busbar. 3. P&C. 4. Civil. 5. Database (SCS). 6. Additional unidentified scope. 7. Commissioning.	GIS Bay (Connection Asset) £5,500k	£5,500k	N/A, we are using a special runway specific for LT29	2034	Complex +	NGET are not offering connection assets.
Templeborough 275kV	AIS Substation & Busbar Extension 1. Extend the busbar. 2. Relocate services and equipment to allow the extension. 3. P&C. 4. Civil. 5. Database (SCS). 6. Additional unidentified scope. 7. Commissioning.	Mesh 'Connection Bay' £5,500k	£6,500k	N/A, we are using a special runway specific for LT29.	2031	Complex	As this is a Mesh Corner Substation there is the requirement of a NGET connection bay. This site will potentially require additional civil works to remove redundant transformer plinth and bund; these costs have not been evaluated at this time
New High Marnham 400kV	AIS Busbar Extension 1. Extend the busbar. 2. P&C. 3. Civil. 4. Database (SCS). 5. Additional unidentified scope. 6. Commissioning.	N/A	£4,000k	N/A, we are using a special runway specific for LT29.	2031	Simple	The assumption is SI build the new substation.

Ocker Hill 275kV	AIS Mesh Corner Extension/ reconfiguration 1. Extend the busbar. 2. Mesh corner Modification 3. P&C. 4. Civil. 5. Database (SCS). 6. Additional unidentified scope. 7. Commissioning.	Mesh 'Connection Bay' £5,500k	£6,800k	N/A, we are using a special runway specific for LT29.	2032	Complex +	Likely to be an unfeasible point of connection due to cost and pipeline of work.
Drakelow 400kV	AIS Busbar Extension 1. Extend the busbar. 2. P&C. 3. Civil. 4. Database (SCS). 5. Additional unidentified scope. 6. Commissioning	N/A	£4,000k	N/A, we are using a special runway specific for LT29.	2031	Simple	The connection within the site will be simple, the routing of the cable will be complex
New Kemsley 400kV	GIS Busbar Extension 1. Extend the busbar. 2. P&C. 3. Civil. 4. Database (SCS). 5. Additional unidentified scope. 6. Commissioning.	GIS Bay (Connection Asset) £5,500k	£5,500k	N/A, we are using a special runway specific for LT29.	2033 With risk of delays.	Complex +	At present there are no approved designs or layouts. NGET are not offering connection assets.
Richborough 400kV	GIS Busbar Extension 1. Extend the busbar. 2. P&C. 3. Civil. 4. Database (SCS). 5. Additional unidentified scope. 6. Commissioning.	GIS Bay (Connection Asset) £5,500k	£5,500	N/A, we are using a special runway specific for LT29.	2029 With risk of delays.	Complex	Richborough is a very difficult substation for the Users to access and gain cable corridors. NGET are not offering connection assets.
Landulph 400kV	AIS Busbar Extension 1. Extend the busbar. 2. P&C. 3. Civil. 4. Database (SCS). 5. Additional unidentified scope. 6. Commissioning	N/A	£4,000k	N/A, we are using a special runway specific for LT29	2030 With risk of delays.	Complex	The site has a non-standard layout and design. Any connection and modification will need to be factored into the design of the connection.
New Woolavington 400kV	GIS Substation & Busbar Extension 1. Extend the building. 2. Extend the busbar. 3. P&C. 4. Civil. 5. Database (SCS). 6. Additional unidentified scope. Commissioning.	GIS Bay (Connection Asset) £5,500k	£5,500k	N/A, we are using a special runway specific for LT29.	2030	Complex	NGET are not offering connection assets.

Any impact on costs requiring either ARS or PoW are as follows:

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

Indicative high level scope cost summary:

Scope Type	Costs
AIS Busbar Extension	£4m
AIS Substation & Busbar Extension	£5.5m
AIS Asset Transfer	£0.25m
AIS Mesh Corner Modifications	£5m
AIS Mesh Corner 'Connection Bay'	£5.5m
GIS Busbar Extension	£3.5m
GIS Substation & Busbar Extension	£5.5m
GIS Asset Transfer	£0.25m
GIS Bay (Connection Asset) *	£5.5m
Point on Wave	£0.15m
Automatic Reactive Switching (New)	£0.40m
Automatic Reactive Switching (Extension)	£0.15m
Tower diversion and OHL works	£1.5m
Cable diversion	£0.3m
	(+/- 25%)

*Note 1: NGET are not offering connection assets.

Note 2: All prices are indicative based on current estimates, actual prices will be built up on a site by site, scheme by scheme basis based on the complexity on each connection.

Scope Complexity Ranking key:

Simple	**Straight forward bay transfer or busbar extension.**
Standard	**Standard busbar extension and / or modifications to the existing substation layout.**
Complex	**Substation extension / building extension including space constraints for cable routes.**
Complex +	**New Substation or in-situ substation re-build / Up rating.**
N/A	**Site not considered further as no connection opportunities or complex work been delivered to meet Stability service date**

Indicative NGET Costs: Additional Details

Infrastructure associated costs and spend profiles – by type of work.

Based on the commercial boundaries as defined for the study, below are indictive infrastructure costs in terms of 'Final Sums' securities liabilities.

Type of work	Oct 25-Mar 26	Apr 26-Sep 26	Oct 26-Mar 27	Apr 27-Sep 27	Oct 27-Mar 28	Apr 28-Sep 28	Oct 28-Mar 29
AIS Busbar Extension	20	60	140	220	420	620	820
AIS Substation & Busbar Extension	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
AIS Asset Transfer	1.25	6.25	18.75	31.25	56.25	93.75	137.5
AIS Mesh Corner Modifications	25	75	175	275	525	775	1025
AIS Mesh Corner 'Connection Bay'	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
GIS Busbar Extension	17.5	52.5	122.5	192.5	367.5	542.5	717.5
GIS Substation & Busbar Extension	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
GIS Asset Transfer	1.25	6.25	18.75	31.25	56.25	93.75	137.5
GIS Bay (Connection Asset)	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
Point on Wave	0.75	2.25	5.25	8.25	15.75	23.25	30.75
Automatic Reactive Switching (New)	2	6	14	22	42	62	82
Automatic Reactive Switching (Extension)	0.75	2.25	5.25	8.25	15.75	23.25	30.75
Tower diversion and OHL works	7.5	22.5	52.5	82.5	157.5	232.5	307.5
Cable diversion	1.5	4.5	10.5	16.5	31.5	46.5	61.5
Activity	Application / Customer Offer	Application / Customer Offer / Optioneering	Optioneering / Development	Development	Development	Development / Delivery	Delivery
Period	Q4/Q1	Q2/Q3	Q4/Q1	Q2/Q3	Q4/Q1	Q2/Q3	Q4/Q1

Cost base £k (+/- 25%).

Type of work	Apr 29-Sep 29	Oct 29-Mar 30	Apr 30-Sep 30	Oct 30-Mar 31	Apr 31-Sep 31	Oct 31- Mar 32	Total Check (£k)
AIS Busbar Extension	1620	2620	3620	3940	3980	4000	4,000
AIS Substation & Busbar Extension	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5,500
AIS Asset Transfer	175	237.5	250	-	-	-	250
AIS Mesh Corner Modifications	2025	3275	4525	4925	4975	5000	5000
AIS Mesh Corner 'Connection Bay'	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5500
GIS Busbar Extension	1417.5	2292.5	3167.5	3447.5	3482.5	3500	3,500
GIS Substation & Busbar Extension	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5,500
GIS Asset Transfer	175	237.5	250	-	-	-	250
GIS Bay (Connection Asset)	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5,500
Point on Wave	60.75	98.25	135.75	147.75	149.25	150	150
Automatic Reactive Switching (New)	162	262	362	394	398	400	400
Automatic Reactive Switching (Extension)	60.75	98.25	135.75	147.75	149.25	150	150
Tower diversion and OHL works	607.5	982.5	1357.5	1477.5	1492.5	1500	1500
Cable diversion	121.5	196.5	271.5	295.5	298.5	300	300
Activity	Delivery	Delivery	Delivery	Delivery / Closure	Closure	Closure	
Period	Q2/Q3	Q4/Q1	Q2/Q3	Q4/Q1	Q2/Q3	Q4/Q1	

Cost base £k (+/- 25%).

Note 1: Costs are not outturned.

Note 2: As part of the Pathfinder solutions there may be the requirement to add additional electronic / control functionality:

1. Point on Wave (PoW) switching capability, and / or
2. Automatic Reactive Switching (ARS) capability.

Function	Description	Cost (£k)
Point on Wave	PoW is controlled switching (either opening or closing) of a CB phase by phase to minimise system disturbance or CB duty mainly on highly reactive circuits i.e. Shunt reactors or MSC's.	150
Automatic Reactive Switching (New)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	400
Automatic Reactive Switching (Extension)	HV voltage is control by switching static reactive plant, Shunt Reactors or MSC's based on system measurements.	150

Cost base £k (+/- 25%).

Note 1: For Mesh Corner substation there may be the requirement to retrofit a Feeder Circuit Breaker Bay if there isn't one already installed at the site. This will be required if the PoC is in the Feeder protection zone and not the Mesh Corner protection zone. This will be determined in the BP500 4.2 Optioneering phase. This will require additional attributable scope. The indicative cost profile for a retrofit Feeder Circuit Breaker Bay.

Indicative spend profile:

Type of work	Oct 25-Mar 26	Apr 26-Sep 26	Oct 26-Mar 27	Apr 27-Sep 27	Oct 27-Mar 28	Apr 28-Sep 28	Oct 28-Mar 29
AIS Mesh Corner 'Feeder Bay'	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
GIS Mesh Corner 'Feeder Bay'	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
Type of work	Apr 29-Sep 29	Oct 29-Mar 30	Apr 30-Sep 30	Oct 30-Mar 31	Apr 31-Sep 31	Oct 31- Mar 32	Total Check (£k)
AIS Mesh Corner 'Feeder Bay'	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5500
GIS Mesh Corner 'Feeder Bay'	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5,500

Cost base £k (+/- 25%).

One-Off Works Costs: More details

Based on a recent review of the criteria for One-Off Works, as defined under CUSC Section 14, Part 1, paragraph 14.4.2, the following activities have been classified as One-Off Works triggered by the User's works:

- Modifications to 'in-delivery' NGET schemes, e.g. flood defence schemes.
- Commissioning Support, in the form of a Commissioning Engineer and Senior Authorised Person.
- NGET Technical Assurance, in the form of a Transmission Engineer (Primary and/or Secondary, as required).

NOTE: The role of the NGET Transmission Engineer is not to "assure/approve/verify" the User's design, but rather to review the User's design solution to ensure it does not impact the overall operation of the site/network when the User's circuit is connected and energised. All assets need to comply with the RES.

- Database change associated with the commissioning and energisation of the User's circuit.
- It includes such activities as database modifications, testing, installation, quality assurance and documentation.

Below is an example of the One-Off Works costs for NGET resources and associated database change, based on the following assumptions:

- No works related to 'in-delivery' NGET schemes triggered
- Completion date 31st March 2028 for a spare bay already in existence.
- NGET resources, as per resource profile below; supporting a 4-week commissioning period prior to Completion Date.

Resource	Duties	Total Working Days
Transmission Engineer	P&C technical review consisting of 1 site meeting per circuit, 1 design review meeting per circuit, drawing reviews	10
Transmission Engineer	Commissioning Officer duties consisting of 1 commissioning panel meeting per circuit, closing meeting and non-meeting works	5
Commissioning Engineer	Non-outage works attendance to 1 design review meeting per circuit: 1 pre-construction site meeting per circuit & non-meeting works	10
Senior Authorised Person (SAP)	Non-outage works attendance to 1 design review meeting per circuit: 1 pre-construction site meeting per circuit & access to NGET site, Work Permit issuing	10
Commissioning Engineer	Support during outage and commissioning period.	18 (3 days per week)
SAP	Support during outage and commissioning period.	12 (2 days per week)

	One-Off Works Costs (£k)
Database Change	£200k
NGET Technical Assurance	£30k
NGET ETAM Ops	£30k
Total	£260k

Mesh Corner Site Information

1. Part of the review was / is to identify potential connection sites and identify potential risks associated with these sites. Several sites have been identified during substation site reviews that have older types of Protection and Control equipment, and software installed that is no longer supported and NGET are unable to expand and re-programme due to the age of the equipment. These types of protection and control equipment are fit for purpose in their present set up and running arrangements but are not suitable to be extended to facilitate a new connection. The risks associated with these sites are:

   - Programme risk, securing required outages to enable equipment replacement, the equipment will need to be replaced,
   - Costs, the costs associated with modification to an existing set up are lower than a compete replacement of the equipment on that section of the substation.

1. Protection and Control / Substation Control System timeline for obsolete equipment:

NGET are developing options with manufacturers to support and update this equipment. NGET's view is that a connection in 2028 will still be achievable, but the solution is still in development.

For Mesh Corner connections where the customer is a generator (exports TEC) any connection to a Mesh Corner is not SQSS compliant. If a customer chooses to connect then the connection will not be firm and there are inherent risks associated with normal operation including fault and planned outages.

- If the customer is treated as demand user, i.e. 0MW TEC exported, they can connect to a Mesh Corner, but this is customer choice and deemed a non-firm connection.
- If there is a fault on the Mesh Corner, then this section will be out of service and the Customer will not be able to generate or provide services. There is a reduced risk on a double busbar configured substation as the connection has the flexibility of been connected to the Main and Reserve busbar which provides an additional level of security.

1. If there is a fault on the Circuit Breaker, then this section of the Mesh Corner will be out of service and the Customer will not be able to generate or provide services. This is not the same on a Double Busbar substation design as a customer is connected to Main and Reserve busbar which provides an additional level of security.

   - Several sites in this report are constructed as Mesh Corner substations.
   - During the substation site review, and learning lessons from previous Stability Pathfinder connections, there is a requirement for the installation of a local CB, CT and VT along with the disconnector at mesh substations. This is to reduce the impact/risk of faults onto the NGET Mesh Corner, as a result of possible long sections of User owned cable, (as may be installed as part of the physical connection to the NGET substation). At the sites where there is a limited amount of space to allow the bay equipment to be installed inside the existing substation compound and in close proximity to the connection point, the use of Disconnecting Circuit Breakers will be identified and accounted for the infrastructure securities spend profile. Where a separate Mesh Corner isolator and CB are capable of being installed only the isolator and its connection to the Mesh Corner will be accounted for in the infrastructure securities spend profile (the works downstream of the isolator will be accounted for in the connection asset securities profile, or if additional to nominal NGET ownership in the One-Off Works securities profile), see Note.

   NOTE on asset classification: NGET will be classifying all assets from the Mesh Corner down to and including the first isolator (or DCB if used in place of separate isolator and circuit breaker) as Infrastructure in line with CUSC 14.2.6b.

   - As the Mesh Corner connection is for a single user, NGET will classify all further downstream assets (to NGET's own nominal ownership boundary) as Connection Assets in the case of connecting a CUSC User in line with CUSC 14.2.6b OR in the case of connecting another STC Party (e.g. a TO) the same assets will be classified as Infrastructure Assets.
   - Where a CUSC User requests an alternate ownership boundary downstream of NGET's own nominal ownership boundary, and subject to NGET's agreement to do so, NGET will classify the additional assets under NGET's ownership as One-Off Works and charge the CUSC User accordingly.

- In the scenario of a connection to a double busbar substation, it is expected by the nature of the physical arrangement of the substation and associated spare bays that there is sufficient space for the user to locate their CB, disconnector and CTs, within the bay area.

At the time of assessing each of the connection applications and subsequent offers NGET will determine whether a Category 1 intertrip is required. The anticipated costs associated with this work is circa £330k.

CB Upgrading / Replacement

As part of the completed power system studies several interventions have been identified whose scope will differ dependent upon the driver behind the need and the type of equipment presently installed at those sites and any future planned works at those sites. The intervention could range from a circuit breaker refurbishment to a full bay replacement including bay equipment.

The intervention could include but is not limited to:

1) Circuit breaker refurbishment / replacement,
2) Voltage Transformer replacement,
3) Current Transformer replacement,
4) Disconnector replacement,
5) Earth Switch replacement,
6) Foundation / base / plinth reinforcement / replacement,
7) Protection replacement,
8) Busbar upgrading.

Each customer connection will need to be treated on its own merits due to the technology type connecting and its impact on the running arrangement and the limits of the equipment installed.

The indicative cost range for the interventions range from £500k to £4m per bay / point of connection, depending on scope of the works and may be substantially more if the whole site replacement is required. These costs would be infrastructure costs.

The programme of works to deliver this reinforcement work will need to be aligned where possible with the customer connection, otherwise where there are fault level issues, a customer cannot connect until this is addressed. The type of intervention and requirements would need to be assessed in more detail during the development of the project, along with confirming outages and resource availability.

Due to the undefined scope of works it is not possible to indicate when the works would be complete. A simple CB replacement may be accommodated in the timescales (2031 service date) if outages / resources are available, but a bay replacement would need additional resource, outage durations, contracting and procurement which may not be available due to other works in the outage plan. A whole substation replacement would take several years to develop and deliver.

There is currently an optimised plan of work for the T2+2 and T3 period, based on the known baseline plus additional work expected through customer connections, ASTI, Net Zero and other requirements.

New plan requirements will be assessed on a case-by-case basis when outage and resource needs can be identified, to ensure prioritisation is allocated to the right work based on several strategic priorities. It is impossible to state at this stage whether our workplan can easily complete these works, due to the unknown scope, outages and resource requirement at this time.

Type of work	Oct 25-Mar 26	Apr 26-Sep 26	Oct 26-Mar 27	Apr 27-Sep 27	Oct 27-Mar 28	Apr 28-Sep 28	Oct 28-Mar 29
AIS 'Upgrade Bay'	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
GIS 'Upgrade Bay'	27.5	82.5	192.5	302.5	577.5	852.5	1127.5
Type of work	Apr 29-Sep 29	Oct 29-Mar 30	Apr 30-Sep 30	Oct 30-Mar 31	Apr 31-Sep 31	Oct 31- Mar 32	Total Check (£k)
AIS 'Upgrade Bay'	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5500
GIS 'Upgrade Bay'	2227.5	3602.5	4977.5	5417.5	5472.5	5500	5,500

NGET Note: This table can be used to understand the securities required where a Tenderer chooses to rely on a reserved bay that triggers attributable circuit breaker upgrades (refer to System Design Review section for more information). The table above provides indicative security values that may be required depending on the substation type that requires the upgrade. Note that these values are indicative (+/- 25%).

System Study Review

The System Design Review Section presents the outcome of the fault level and voltage performance assessments that were carried out for all the reserved bays at the substations where NESO identified the need of the Pathfinder Feasibility Study.

Thermal (MW) Review

NESO also requested that thermal headroom to be provided for each site where available.

However, the current contracted generation queue means that there isn't any MW (thermal) capacity available on the transmission network prior to significant transmission reinforcements being completed. This is the case for the reserved bay substations in all regions considered as reserved bay connection points as part of the LT29 tender. As such, for the purpose of this report, thermal headroom has not been considered any further.

The construction planning assumptions (CPA) that NESO provided to enable these studies were based on the generation queue up to 2029, whereas the current generation connections queue extends to connections of capacity, and associated connection dates well beyond this date. As a result, a study carried out to determine thermal headroom based on a 2029 scenario would not present a realistic view of actual available headroom. In addition, the Construction Planning Assumptions (CPA's) presented by NESO were based on summer minimum demand, and Winter peak demand scenarios. Whilst the Winter peak demand scenario can be used to determine thermal headroom, the summer minimum demand scenario is wholly unsuitable for determining thermal headroom as the loading of transmission apparatus under this scenario is typically well below the loadings that would occur during Seasonal peak demand scenarios when a greater volume of generation plant would be despatched.

Assumptions (Fault Level and Voltage)

When applying the Summer CPA, difficulties were encountered in establishing a converging load flow whilst trying to remain wholly consistent with the assumptions that NESO had provided. The table summarises all items where there has been a divergence from NESO's assumptions.

Study Type	Network Model Background Assumption
Voltage Study	The planning assumptions provided by NESO were incorporated into the NGET network model. However, the following modelling deviations were required for the summer minimum demand scenario: • No curtailment was considered in the shunt reactor ratings, in accordance with ETYS. • In line with BP1202, all Voltage Control Circuits (VCCs) were modelled as switched out, except for the Pembroke to Walham circuit. • All Quadrature Boosters (QBs) were set to neutral tap, except for Legacy QB4. • Additional reactive power support considered from Wind generation • The following assumptions were used for the Winter peak demand scenario: ▫ According to NESO Background, the total generation (E&W + B6 Flow) is ~44.5GW, while the demand is ~39.1 GW. After accounting for 2 GW transmission losses, the remaining generation imbalance managed using External HVDC interconnections.

	▫ For the analysis of the North region substations, South region HVDC links were utilised, and vice versa for the South region substations. The assumptions made were intended to ensure that they do not impact the substation under study.
Fault Level	All 275 kV and 400 kV generators were modelled as in service, in accordance with the NESO-provided winter background scenario. The system demand was maintained at approximately 65 GW. Note: Only generators listed in the NESO-provided 2029 Winter Peak scenario were considered, not all generators contracted up to 2029 were included in the assessment.

Fault level headroom has been provided as requested by NESO, however it should be noted that the issue concerning the contracted generation queue, as described in the paragraph above, is also relevant for fault level studies. As such the fault level headroom information presented below is indicative and only reflects the generation that is contracted to connect up to 2029.

Greystones A & B 275kV

Fault Level Analysis

Context

The SCL injection was modelled by using a static generator with the following parameters and value:

Sub-Transient SCL = 6416.5 MVA or 13.47 kA
Transient SCL = 6416.5 MVA or 13.47 kA
X/R ratio of 10.

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Greystone A/B 275kV	1100 MVA	Sub-Transient: 6416.5 MVA Transient: 6416.5 MVA

Conclusion

At Greystones A & B 275kV, there were no fault-level constraints observed based on the current network configuration and study assumptions. Refer to the result table for the available SCL headroom at Greystones A & B 275kV. The limiting factor is three phase fault capability of S10, P20 & W30 circuit breakers.

Voltage Analysis

Context

At Greystones A 275kV Substation, -33 MVAr of inductive reactive power and +33 MVAr of capacitive reactive power have been allocated to the reserved bay. These values were analysed against the summer minimum and winter maximum background conditions, respectively.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Greystone A/B 275kV	-33 MVAr	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Greystone A/B 275kV	At least 33 MVAr	-33 MVAr	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Greystone A/B 275kV	At least 33 MVAr	-33 MVAr	NO <6%

Conclusion

The Target MVAr headroom of 33 MVAr (absorption) has been achieved without any Voltage step issues. The pre-fault steady state and Steady state voltage compliance were achieved for the target MVAr.

Winter Maximum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Greystone A/B 275kV	+33 MVAr	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Greystone A/B 275kV	At least +33 MVAr	+50 MVAr	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Greystone A/B 275kV	At least +33 MVAr	+50 MVAr	NO <6%

Conclusion

The Target MVAr headroom of 33 MVAr (injection) has been achieved without any Voltage step issues. The pre-fault steady state and Steady state voltage compliance were achieved for the target MVAr.

Penwortham 400kV

Fault Level Analysis

Context

The available headroom for Penwortham 400kV substation was peak as shown in the table below

- Sub-Transient SCL = 2078MVA or 3 kA
- Transient SCL = 1386MVA or 2 kA
- X/R = 10

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Penwortham 400kV	TBD by studies	Sub-Transient: 2078MVA Transient: 1386MVA

Conclusion

At Penwortham 400kV, there were no-fault-level constraints observed based on the current network configuration and study assumptions. Refer to the result table for the available SCL headroom at Penwortham 400 kV.

Voltage Analysis

Context

Penwortham 400kV substation results for Summer Minimum scenario and Winter Maximum scenario.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual inductive reactive power provider before the step change limit was breached.

Please note that given the network conditions it was not possible to set the voltages within SQSS limits at all substations

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Penwortham 400kV	- 33	200MVAr

H3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Penwortham 400 kV	-170	200MVAr	NO

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual inductive reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Penwortham 400kV	-170	200MVAr	NO

Conclusion

The Target MVAr headroom of –170 MVAr (absorption) has been achieved (-200MVAr available) without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were achieved for the target MVAr.

Please note that pre-fault steady-state voltage levels comply with the limits defined by the SQSS MVAr injection headroom.

Winter Maximum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of a capacitive reactive power provider could be before the limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Penwortham 400kV	+33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Penwortham 400kV	170 MVAr	+200 MVAr	No <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Penwortham 400kV	+170 MVAr	+200 MVAr	NO <6%

Conclusion

In Winter Scenario, The Target MVAr headroom of +170 MVAr(absorption) has been achieved (+200 MVAr Available) without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were achieved for the target MVAr. This is in compliance with the SQSS.

Templeborough 275kV

Fault Level Analysis

Context

The SCL injection was modelled using a static generator with the following parameters and value:

- Sub-Transient SCL = 2391.6 MVA or 5.02 kA
- Transient SCL = 2391.6 MVA or 5.02
- X/R ratio of 10.

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Templeborough 275kV	TBD by studies	Sub-Transient: 2391.6 MVA Transient: 2391.6 MVA

Conclusion

At Templeborough 275kV, there are no fault-level constraints observed based with the current network configuration and study assumptions. Refer to the result table for the available SCL headroom at Templeborough 275kV. The limiting factor is the three phase fault capability of S10 & S20 circuit breakers.

Voltage Analysis

Context

At Templeborough 275kV Substation, -150 MVAr of inductive reactive power and +130 MVAr of capacitive reactive power have been allocated to the reserved bay. These values were analysed against the summer minimum and winter maximum background conditions, respectively.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached. Please note that the pre-fault steady state upper limits were breached in NESO's summer minimum 2029 background.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Templeborough 275kV	-33 MVAr	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Templeborough 275kV	-129 MVAr	-150 MVAr	No <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the devices connected to the system via a single circuit breaker.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Templeborough 275kV	-129 MVAr	-150 MVAr	NO <6%

Conclusion

The Target MVAr headroom of 129 MVAr(absorption) has been achieved without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were achieved for the target MVAr.

Please note that the starting voltage level before considering reactive power is non-compliant, with voltage level of 292.5kV.

Winter Maximum Scenario

Results

Several scenarios were evaluated to determine the maximum MVAr capacity of an individual reactive power provider could be before the limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Templeborough 275kV	+33 MVAr	YES

H3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Templeborough 275kV	+129 MVAr	+130 MVAr	No <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple devices were run in parallel, the less onerous limit of 6% can be used to as a limit for the loss of all devices connected via a single circuit breaker.

H6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Templeborough 275kV	+129 MVAr	+130 MVAr	NO <6%

Conclusion

The Target MVAr headroom of 129 MVAr(injection) has been achieved without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were achieved for the target MVAr.

New High Marnham 400kV

Fault Level Analysis

Context

The SCL injection was modelled using a static generator with the following parameters and value:

- Sub-Transient SCL = 2566.6 MVA or 5.39 kA
- Transient SCL = 2566.6 MVA or 5.39 kA
- X/R ratio of 10.

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
New High Marnham 400kV	TBD by studies	Sub-Transient: 2566.6 MVA Transient: 2566.6 MVA

Conclusion

At the New High Marnham 400kV substation, there are no fault-level constraints observed based on the current network configuration and study assumptions. Refer to the result table for the available SCL headroom at New High Marnham 400 kV. The limiting factor is three phase fault capability of X230 & X130 circuit breakers.

Voltage Analysis

Context

At New High Marnham 400kV Substation, -150 MVAr of inductive reactive power and +150 MVAr of capacitive reactive power have been allocated to the reserved bay. These values were analysed against the summer minimum and winter maximum background conditions, respectively.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached. Please note that the pre-fault steady state upper limits were breached in NESO's summer minimum 2029 background.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
New High Marnham 400kV	-33 MVAr	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
New High Marnham 400kV	-101 MVAr	-150 MVAr	No <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
New High Marnham 400kV	-101 MVAr	-150 MVAr	NO <6%

Conclusion

The Target MVAr headroom of 101 MVAr(absorption) has been achieved without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were achieved for the target MVAr.

Please note that the starting voltage level before considering the reactive power support is non-compliant, with a voltage level of 420kV.

Winter Maximum Scenario

Results

Several scenarios were evaluated to determine what the maximum capacity of an individual reactive power provider could be before the limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
New High Marnham 400kV	+33 MVAr	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
New High Marnham 400kV	+101 MVAr	+150 MVAr	No <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of both devices connected via a single circuit breaker.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
New High Marnham 400kV	+101 MVAr	+150 MVAr	NO <6%

Conclusion

The Target MVAr headroom of 101 MVAr(injection) has been achieved without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were achieved for the target MVAr.

Ocker Hill 275kV

Fault Level Analysis

Context

The available headroom for Ocker Hill 275kV:

Sub-transient – 5800 MVA or 12.17 kA

Transient - 5382 MVA or 11.3 kA

X/R ratio of 10

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Ocker Hill 275 kV	Maximum available	Sub-transient: 5800 MVA Transient: 5382 MVA

Conclusion

No required reinforcements to achieve the SCL headroom. The limiting factor is 3 phase fault capability of S10, S20, S30 & S40 circuit breakers.

Voltage Analysis

Context

Ocker Hill 275kV substation results for Summer Minimum 2029 scenario and Winter Maximum 2029 scenario.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual inductive reactive power provider was before the step change limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Ocker Hill 275kV	-33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Ocker Hill 275kV	-120	-130	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual inductive reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Ocker Hill 275kV	-120	-230	NO <6%

Conclusion

The target of 120 MVAr (inductive reactive provider) can be achieved with a step change voltage compliance. Pre-fault steady state and steady state (post-fault) voltage compliance were not achieved for the target MVAr.

Winter Maximum Scenario

Results

Several scenarios were evaluated to determine the maximum capability of an individual capacitive reactive power provider before the step change limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Ocker Hill 275kV	+33	0

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Ocker Hill 275kV	+120	0	Yes <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual capacitive reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Ocker Hill 275kV	+120	0	Yes <6%

Conclusion

The minimum target of 33 of capacitive MVAr cannot be achieved due to high voltage and non-convergence in the Winter Peak 2029 study model. Reinforcements to facilitate the minimum MVAr requirements could not have been determined due to complexity and the nature of the voltage profile of the model.

Drakelow 400kV

Fault Level Analysis

Context

The available headroom for Drakelow 400kV:

Sub-transient – 3100 MVA or 4.47 kA

Transient – 2600 MVA or 3.75 kA

X/R ratio of 10

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Drakelow 400kV	Maximum available	Sub-transient: 3100 MVA Transient: 2600 MVA

Conclusion

At Drakelow 400kV, there are no fault-level constraints observed based with the current network configuration and study assumptions. Refer to the result table for the available SCL headroom at Drakelow 400kV. The limiting factor is the three phase fault capability of infrastructure equipment, currently rated for 50kA.

Voltage Analysis

Context

Drakelow 400kV substation results for Summer Minimum 2029 scenario and Winter Maximum 2029 scenario.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual inductive reactive power provider before the step change limit was breached. The scenarios were evaluated as per NESO's Summer Minimum 2029 scenario provided to NGET.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Drakelow 400kV	-33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Drakelow 400kV	-175	-825	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual inductive reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Drakelow 400kV	-175	-1525	NO <6%

Conclusion

The target of -175 MVAr (inductive reactive provider) can be achieved whilst remaining within a voltage step change voltage compliance. Steady state (post fault) voltage compliance was not achieved for the target MVAr.

Winter Maximum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual inductive reactive power provider before the step change limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Drakelow 400kV	+33	Yes

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Drakelow 400kV	+175	+275	No <3%

The banking of multiple capacitive reactive power providers was then considered. Whilst the switching of an individual capacitive reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Drakelow 400kV	+175	+325	No <6%

Conclusion

The target of +175 MVAr (capacitive reactive provider) can be achieved whilst remaining within a voltage step change compliance.

New Kemsley 400kV

Fault Level Analysis

Context

New Kemsley B (north Kent connection node B) 400kV substation is a double turn in from the Grain to Kemsley overhead line , winter background 2029 is considered for the fault level study and all the generators which are connecting up to 2029 have been considered in the background, nearby substations Grain 400kV, Kemsley 400kV, Tilbury 400kV and Kingsnorth 400kV and Wallend (Grain west)400kV substations need to be run in a split running arrangement to mitigate the fault level issues in the area.

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Kemsley B 400kV (north Kent connection node A)	TBC	No available headroom

Conclusion

Fault level Issues were observed at Kemsley B 400kV substation. During the study, Kemsley B was operated in a solid busbar configuration, under this arrangement, it was observed that all circuit breakers at Kemsley B experienced fault levels exceeding their design rating. During this scenario Tilbury 400kV substation which was running in solid mode and acting as a major fault current source. At the same time, Grain 400kV substation was configured in a solid, and Kingsnorth 400kV was running in vertical split but still all the breakers at New Kemsley were breaching.

However, we are seeing fault level issues at Kemsley B during the winter 2029 background.

Voltage Analysis

Context

At the Kemsley B 400kV Substation, a new GIS bay will be allocated for the LT29 project with a maximum of 200MVAr capacitive reactive power and a maximum of -400 MVAr inductive reactive power. Various operational scenarios were considered to assess potential voltage step change issues resulting from switching operations. These reactive power values were analysed against seasonal background conditions: summer minimum 2029 and winter maximum 2029 respectively.

Summer Minimum Scenario

Results

Several cases were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached. Please note that the pre-fault steady state voltage upper limit as per the SQSS were breached in NESO's summer minimum 2029 background.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Kemsley B 400kV (north Kent connection node A)	-33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Kemsley B 400kV (north Kent connection node A)	- 200	- 400	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Kemsley B 400kV (north Kent connection node A)	- 200	- 400	NO <6%

Conclusion

The Target MVAr headroom of –200MVAr (absorption) has been achieved (- 400 MVAr headroom available) without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were not achieved for the target MVAr.

Please note that before connecting the reactive MVArs pre-fault steady-state voltage levels exceed the acceptable upper voltage limits defined by the SQSS and the study results for pre fault steady state voltage is 1.029 p.u. however, after connecting the reactive MVAr voltage levels are within the acceptable range.

The MVAr injection headroom was not assessed in summer minimum scenario as part of this study.

Winter Maximum Scenario

Results

Several scenarios were analysed to determine the upper limit of capacitive reactive power injection capability for a single bay. The assessment was conducted under intact (pre-fault) steady-state voltage conditions, which were confirmed to be compliant with the voltage limits specified in the SQSS, as referenced in NESO's Winter Maximum 2029 background. The range of capacitive reactive power considered for evaluation spanned from a minimum of 33 MVAr to a maximum of 200 MVAr.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Kemsley B 400kV (north Kent connection node A)	+33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Kemsley B 400kV (north Kent connection node A)	+ 200	+200	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Kemsley B 400kV (north Kent connection node A)	+ 200	+200	NO <6%

Conclusion

The study confirmed that 200 MVAr of capacitive reactive power injection under Winter Maximum 2029 conditions ensured full compliance with SQSS Chapter 6 voltage performance criteria, with voltage levels remaining within acceptable limits in both pre-fault and post-fault scenarios in Winter Maximum 2029 system background.

Note 1: it has been noted that de-energising the VCC (Voltage Control Compensation) circuits similar to the operational approach applied under the summer background scenario results in a reduction of regional voltage levels.

Richborough 400kV

Fault Level Analysis

Context

The SCL injection was modelled by using a static generator with the following parameters and value: the total Fault level headroom available at the substation is as below

- Sub-Transient SCL = 1927 MVA or 2.78 kA
- Transient SCL = 1285 MVA or 1.85 kA
- X/R ratio of 10.

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Richborough 400kV	TBC	Sub-Transient: 1927 MVA Transient: 1285 MVA

Conclusion

No Fault level issues were observed at Richborough 400kV but injecting more fault infeed at the substation resulted in more fault level issues at nearby substations. There are existing fault level issues at nearby substation Kemsley 400kV, Kemsley B 400kV and Kingsnorth 400kV, Grain, Grain West and Tilbury 400kV

Voltage Analysis

Context

At Richborough 400kV Substation, two reactive compensation bays rated at -33 MVAr (inductive) and two bays rated at +33 MVAr (capacitive) were modelled and assessed under summer minimum and winter maximum system background conditions respectively. Both configurations were studied for voltage compliance with the relevant SQSS criteria.

Summer Minimum Scenario

Results

Several cases were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached, no breach of the limits was seen in the steady state.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Richborough 400kV	- 33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Richborough 400kV	2 x- 33	2 x - 150	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Richborough 400kV	2 x - 33	2 x - 150	NO <6%

Conclusion

The Target MVAr headroom of 2 bays of -150 MVAr each (absorption) has been achieved without any Voltage step issues. The Pre-fault steady state and post-fault voltage compliance was achieved for the targeted MVAr.

The before and after connecting the reactive MVAr pre-fault steady-state voltage level is within the acceptable limits defined by the SQSS in chapter 6.

Note: The MVAr injection headroom was not assessed as part of summer 2029 background.

Winter Maximum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Richborough 400kV	+ 33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Richborough 400kV	2 x + 33	2 x + 60	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Richborough 400kV	2 x + 33	2 x+ 60	NO <6%

Conclusion

Under Winter Maximum 2029 conditions and NESO study background, the target capacitive reactive power headroom of 120 MVAr was achieved via two 60 MVAr bays. System voltages remained within SQSS Chapter 6 limits in both pre-fault and post-fault scenarios, confirming compliance and allowing for stable reactive power injection to support system performance.

Note 1: However, it has been noted that de-energising the VCC (Voltage Control Compensation) circuits similar to the operational approach applied under the summer background scenario results in a reduction of regional voltage levels.

Landulph 400kV

Fault Level Analysis

Context

The SCL injection was modelled by using a static generator with the following parameters and value: The fault level headroom at Landulph was assessed by modelling a static generator at the MC3 busbar, as connections at MC1 and MC2 are unavailable due to stability concerns and limited space near existing plant. The study determined that the total available fault level headroom at MC3 is 4.49 kA sub-transient and 2.99 kA transient.

- Sub-Transient SCL = 3117 MVA or 4.49 kA
- Transient SCL = 2078 MVA or 2.99 kA
- X/R ratio of 10.

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
Landulph-MC3	TBC	Sub-Transient: 3117 MVA Transient: 2078 MVA

Conclusion

At Landulph 400kV, no fault-level constraints were observed based on the current network configuration and study assumptions. Refer to the result table for the available SCL headroom at Landulph 400kV.

Voltage Analysis

Context

At Landulph 400kV Substation, 2 bays of 150 MVAr each are considered for the reserved bay and studied against summer minimum and winter maximum background.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached. Please note that the pre-fault steady state voltage upper limits as per the SQSS were breached in NESO's summer minimum 2029 background.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Landulph 400kV	-33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Landulph 400kV	2 x -150	2 x -150	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Landulph 400kV	2 x -150	2 x -150	NO <6%

Conclusion

The Target MVAr headroom of - 300 MVA r(absorption) in two bays of -150MVAr each has been achieved without any Voltage step issues. The Pre-fault steady state and steady state (post-fault) voltage compliance were not achieved for the target MVAr.

Please note that the results before connecting the inductive reactive MVAr's (Absorption) the pre fault steady state voltage is 1.059 PU which exceeds the pre-fault steady-state voltage upper-level limit defined by in SQSS chapter 6. Following the connection of the 2 bays of -150MVAr each the voltages are within the voltage limit defined by SQSS chapter 6 and no voltage step change breaching was observed.

Note: The MVAr injection headroom was not assessed as part of this study with summer minimum background.

Winter Maximum Scenario

Results

Several cases were evaluated to determine what the maximum capability of an individual capacitive reactive power provider could be before the limit was breached. Please note that the pre-fault steady state or intact voltage is within the limit as per the SQSS in NESO's winter maximum 2029 background.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Landulph 400kV	+33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Landulph 400kV	2 x +150	2 x +150	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Landulph 400kV	2 x +150	2 x +150	NO <6%

Conclusion

The capacitive MVAr headroom of +300 MVAr(injection) across two bays of +150MVAr each has been achieved without any Voltage step issues with the target 300 MVAr the system voltage will be within the limit. The post-fault voltage compliance was not achieved for the target MVAr because during postfault voltage levels were above the SQSS chapter 6 voltage limits. But no voltage step issue was observed.

Please note that after connecting the capacitive MVAr's (injection) during the winter post-fault the voltage levels are exceeding the acceptable upper limits defined by the SQSS chapter 6 however, before connecting the capacitive MVAr the voltages are within the allowed limits as per the SQSS.

New Woolavington 400kV

Fault Level Analysis

Context

The SCL injection was modelled by using a static generator with the following parameters and value:

- Sub-Transient SCL = 3117 MVA or 4.49 kA
- Transient SCL = 2078 MVA or 2.99 kA
- X/R ratio of 10.

Results

Reserved Bay	NESO Target SCL MVA	Available fault-level (SCL) headroom
New Woolavington-RB1	TBC	Sub-Transient: 3117 MVA Transient: 2078 MVA

Conclusion

At New Woolavington 400kV, no fault-level constraints were observed based on the current network configuration and study assumptions. Refer to the result table for the available SCL headroom at Woolavington 400kV.

Voltage Analysis

Context

At Woolavington 400kV Substation, 276 MVAr is considered for the reserved bay and studied against summer minimum and winter maximum background.

Summer Minimum Scenario

Results

Several scenarios were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached. Please note that the pre-fault steady state voltage upper limits as per the SQSS were breached in NESO's summer minimum 2029 background.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
Woolavington 400kV	- 33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Woolavington 400kV	- 276	-1200	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
Woolavington 400kV	-276	-1200	NO <6%

Conclusion

The Target MVAr headroom of –276MVAr(absorption) has been achieved (-1200 MVAr(absorption) headroom available) has been achieved without any Voltage step issues.

Please note that pre-fault steady-state or intact system voltage levels before connecting reactive MVAr's (Absorption) exceed the acceptable limits defined by the SQSS. After connecting the Reactive MVars the voltages are within the voltage limit defined by SQSS chapter 6.

Note: The reactive MVAr injection headroom was not assessed as part of the summer minimum study.

Winter Maximum Scenario

Results

Several cases were evaluated to determine what the maximum capability of an individual reactive power provider could be before the limit was breached. Please note that the pre-fault steady state voltage is within the limits as per the SQSS for NESO's winter Maximum 2029 background.

Minimum Voltage Assessment

Reserved Bay	NESO Minimum MVAr	Minimum MVAr Available
New Woolavington 400kV	+ 33	YES

H_3 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
New Woolavington 400kV	+ 276	+276	NO <3%

The banking of multiple reactive power providers was then considered. Whilst the switching of an individual reactive power provider must remain under the 3% limit, if multiple were run in parallel, the less onerous limit of 6% can be used to judge the compliance of the loss of the entire facility.

H_6 assessment

Reserved Bay	NESO Target MVAr	Available MVAr Headroom	Voltage Step Issue
New Woolavington 400kV	+276	+276	NO <6%

Conclusion

A capacitive reactive power injection headroom of +276 MVAr was achieved without any voltage step issues. pre-fault steady-state voltages remained within acceptable thresholds prior to injection; the post-fault steady-state voltages surpassed the scenario-defined limits following the injection.

Glossary

Abbreviation	Definition
PoW	Point on Wave
PoC	Point of Connection
ARS	Automatic Reactive Switching
AIS	Air Insulated Switchgear
GIS	Gas Insulated Switchgear
GIL	Gas Insulated Line
NETS	National Electricity Transmission System
NGESO	National Grid Electricity System Operator
NGET	National Grid Electricity Transmission
SCL	Short Circuit Level
SQSS	Security and Quality of Supply Standard
SI	Strategic Infrastructure